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Development of technical video tutorials using the Triple Diamond design process model

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Statutory Declaration

I hereby declare that I am the sole author of the thesis entitled *Development of technical video tutorials using the Triple Diamond design process model*. I have not used sources or means without declaration in the text. Any thoughts from others or literal quotations are clearly marked.

In Prague on the 12th of December, 2022

.....

Anna Manukian

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Abstract

This bachelor's thesis is focused on product development using the Triple Diamond design process model. The developed product is primarily aimed at VŠE students and describes the steps involved in accessing VŠE VPN and NAS while also giving an overview of those technologies. The chosen media is a video with a voice-over.

The theoretical part is concentrated on Human-centred design and its best practices, different frameworks and tools and particularly explores the Double and Triple Diamond design process models. Then a typology of educational video is discussed, followed by an explanation of cognitive overload, multimedia usage in video and accessibility. Finally, the theoretical part covers informational technology essential for product development.

The practical part describes project development using the Triple Diamond design process model from the early stages which include research, problem definition and solution discovery to concept validation and development consisting of production and post-production.

The outcome of this project is a set of 10 accessible videos about the VŠE VPN and KME NAS which proved to solve all the users' problems discovered in this thesis. After watching the produced videos, users performed two times better on a test on the school VPN and NAS. The needed services were accessed faster than expected and the videos were sufficient for solving the few errors which occurred during the testing. The user testing participants generally agreed or strongly agreed with positive statements about the videos and described them as informative, useful, understandable, concise, engaging, and visually appealing.

Finally, advice on distribution and possible topics for other manuals is provided.

Keywords: HCD, triple diamond design process model, VPN, educational video, accessible video

Abstrakt

Tato bakalářská práce je zaměřena na vývoj produktu s využitím modelu vývoje Triple Diamond. Vyvinutý produkt je primárně určen pro studenty VŠE a popisuje kroky potřebné pro připojení na školní VPN a NAS a zároveň poskytuje přehled těchto technologií. Zvoleným médiem je video s komentářem.

Teoretická část je zaměřena na Design zaměřený na člověka a jeho osvědčené praxe, různé frameworky a nástroje a zejména zkoumá Double a Triple Diamond design systém. Poté je diskutována typologie vzdělávacího videa, následuje vysvětlení kognitivního přetížení, využití multimédií ve videu a dostupnost. Dále v teoretické části jsou popsány informační technologie nezbytné pro vývoj produktu.

Praktická část popisuje vývoj projektu s využitím Triple Diamond od raných fází, které zahrnují výzkum, definici problému a hledání řešení až po validaci konceptu a vývoj sestávající z produkce a postprodukce.

Výsledkem tohoto projektu je sada 10 přístupných videí o VPN VŠE a NASu KME, která dokázala vyřešit všechny problémy uživatelů objevené v této práci. Po zhlédnutí vytvořených videí si uživatelé vedli dvakrát lépe v testu ze školního VPN a NASu.

Na VPN a NAS se uživatelé připojili rychleji, než se očekávalo, a videa byla dostatečná pro vyřešení několika chyb, které se během testování vyskytly. Účastníci uživatelského testování obecně souhlasili nebo rozhodně souhlasili s pozitivními výroky o videech a charakterizovali je jako informativní, užitečná, srozumitelná, stručná, poutavá a vizuálně přitažlivá.

Na závěr jsou uvedeny rady týkající se distribuce a možné témata pro další manuály.

Klíčová slova: HCD, triple diamond, VPN, edukační videa, přístupná videa

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1. Introduction

At The Prague University of Economics and Business (VŠE), there are more than 13 000 students, approximately 600 academic workers, and a number of external lecturers. [40] All of them use some VŠE services like Intranet, InSIS, Office365, VMware Horizon, Azure Dev Tools, and others. Some software can only be used on certain computers where it has been physically installed. To access certain services like the intranet, InSIS, or Office 365, it is enough to log in with school credentials. [60] Still, some services can only be accessed from the school network, so you need a Virtual Private Network (VPN) connection to the school network to get to them. To connect to the Multimedia department (KME) Network Attached Storage (NAS), for example, the device must be connected to either the VŠE network or the schoolVPN. [43] The school VPN may also be used to access all school services safely if connected to an unsecured network.

The main objective of this work was to create tutorials on a topic that concerned students in a convenient format for them. That goal was to be reached with the help of Human-centered design (HCD); this project was conducted in accordance with the triple diamond design process model. During this project, it was discovered that there was a demand for explainer videos about VPN and NAS and practical tutorials on how to access them from different platforms.

1.1 Thesis structure

The theoretical part is divided into three main sections. The first section concentrates on human-centred design with an emphasis on design process models. In the second part, typologies of educational videos and important concepts regarding their creation are discussed. The last part focuses on informational technologies, namely VPN, NAS, operating systems, and software used to create the videos.

The practical part describes product creation using the triple diamond design process model. The project was put through two tests, validations with experts on the video topics, as well as quantitative and qualitative research.

The outcome of this project is a complete product ready for launch, which is a set of 10 videos in the form of screencasts or animations followed by a voiceover. The success of the project is measured by comparing expected Key Performance Indicators (KPIs) with the real user results in a usability test.

2. Human-Centred Design

As Dan Norman writes in his book *The Design of Everyday Things* [39], HCD “is an approach that puts human needs, capabilities, and behaviour first, then designs to accommodate those needs, capabilities, and ways of behaving.”

To create a good design, it is crucial to understand the users’ psychology. It is needed in order to communicate understandably, especially when something goes wrong, and the user is frustrated. To do so, when something goes wrong, it is important to highlight the solution to the problem so that the user understands the issue and how to overcome it.

2.1 Design thinking

The core of Design Thinking is **solving the right problem**. That means that before designing a solution, first, the real issues need to be discovered. The right solution to a wrong problem will not solve the issue, and therefore the desired result will not be achieved.

In order to organise the design process effectively, different frameworks were created, one of them being the **Double Diamond design process model**.

2.2 The Double Diamond design process model

Developed by the Design Council in 2005, the Double Diamond design process model is “a simple way to describe the steps taken in any design project, irrespective of methods and tools used.” [24]

It is divided into two major steps which are illustrated by the two diamonds:

1. Finding the right problem.
2. Finding the right solution.

Each diamond is then divided into two phases.

1. The Divergence phase is the first phase of each step (diamond). In this phase, the amount of processed data is extended.
2. The Convergence phase concludes each step. In this phase, the ambiguity is reduced by focusing on the most important data.

The entire process is divided into four main stages: **Discover**, **Define**, **Develop** and **Deliver**.

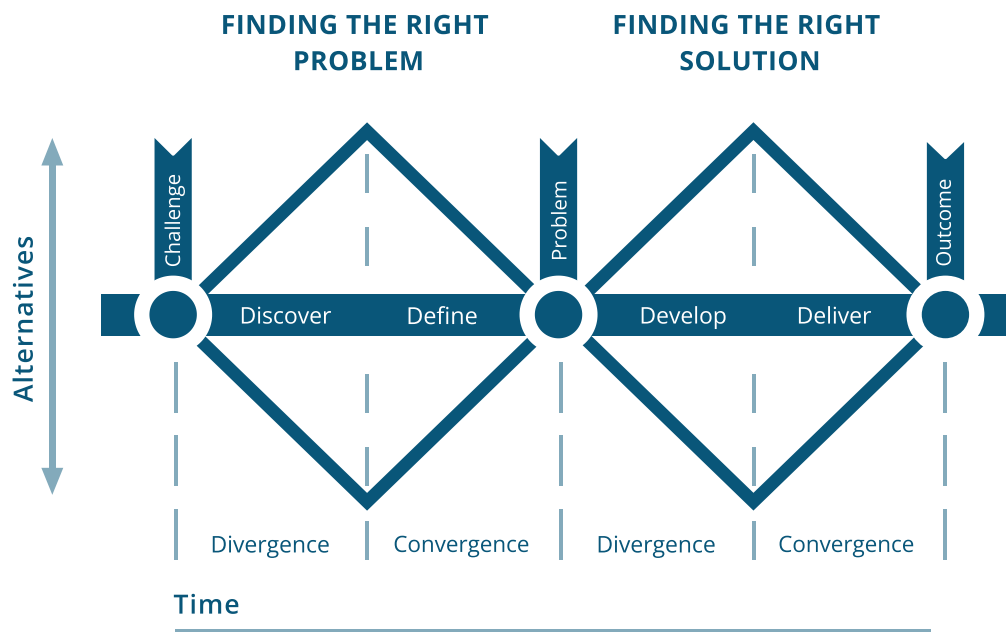


Figure 2.1: The Double Diamond, Manukian, 2022

In the **Discover** phase, the scope of the problem is extended, which is primarily done by performing research. The main goal of this step is to gather as much relevant data as possible. To achieve the desired results, often desk research is not enough, and therefore primary research, often a quantitative one, is carried out. Usually, the aim of this research is to determine, whether there is definitely a need for the discussed project.

Usually, at this stage, **Target audiences** are already known. Once they are determined, User personas can be created. **User personas** are fictional people who represent target audiences' goals, needs, frustrations, and other characteristics (e.g., age, geographical location, income, job position, personality, and gender). They are made to show who the product is meant for so that designers can think of a specific person while working on a project.

The main advantage of this method is **building empathy**: it may be easier to visualise users' needs if you can imagine them as real people rather than a set of characteristics of the target audience. A good User persona reflects the target audience in every aspect. [26]

The following **Define** phase focuses on articulating the problem, thus reducing ambiguity. A good **insight**, which should have emerged in the prior phase, should help formulate the challenge correctly. At the end of this phase, designers have an understanding of what issue they are going to address in the next phase. This issue is usually formulated in the form of a **Problem statement**. [27] One of the possible ways to write it is the following:

[User name] **is a/an** [user characteristics] **who needs** [user need] **because** [insight].

Peter is a traveller who needs an offline map because his internet connection is unstable.

The **Develop** phase starts with ideation. Designers try to give as many possible answers to the problem as possible so that later those ideas can be sorted and assessed. To do so, various methods are used, one of which is **How might we? (HMW)** questions.

How Might We questions

This method helps designers discover various possible solutions. Once the problems are articulated, they are reformulated into questions, which gives the possibility to think of various creative solutions. Those questions should be broad enough to inspire to discover multiple solutions and specific enough to let designers solve only one problem at a time. For instance, “How might we create a good video?” would not serve as a good HMW question because “good” is too broad of a word. That can lead to two problems:

- firstly, designers might have a different idea of what a good video should be like;
- secondly, this broad word does not give any specific insight into the users’ problem.

Customer Journeys

Next, Customer Journeys can be created. They reflect every stage of users’ engagement with the product. There are different frameworks for creating Customer Journeys, some of them being See–Think–Do–Care (STDC) and Awareness–Interest–Desire–Action (AIDA).

Both these frameworks contain an Awareness phase (See in STDC) when the user discovers the product, and an Action (Do in STDC). Step Think in STDC is presented by the Interest and the Desire phases in AIDA, leaving step Care in STDC the main difference between those two frameworks. This step introduces aftercare for the user or customer by providing them with benefits after they have completed the desired action. [7]

Depending on the project, other sensible steps may be added. [37] Those may be:

1. Research (solution is searched, also referred to as See).
2. Consideration (user weights the options, also referred to as Think).
3. Purchase (or other desired action, also referred to as Do or Action).
4. Support (also referred to as Care).
5. Promotion (the user is sharing their experience, both good and bad).

STDC	See		Think		Do	Care	
AIDA	Awareness		Interest	Desire	Action		
Other	Research	Discovery	Consideration		Use	Support	Promotion

Table 2.1: Customer Journey types

Customer Journeys portray a best-case scenario of the target audience (sometimes presented by a user persona) following a **happy path** which means a situation when everything happens according to the original plan and the user was happy with the experience. Nonetheless, it is rarely the case. Therefore, it is important to make designs keeping possible errors in mind. Also, not all the users are going to share the exact characteristics of the target audience. Subsequently, it is necessary to design for **edge cases**, situations which are less likely to occur while using the product, but nonetheless possible. [34]

In the **Deliver** phase several **low-fidelity prototypes** (also referred to as lo-fi prototypes) are created and tested with users. After that, the best solution is chosen and then improved into a **high-fidelity prototype** (also referred to as a hi-fi prototype). Sometimes there are multiple good solutions which are combined after the user testing. [24]

2.3 The Triple Diamond design process model

In 2020, the American company Zendesk, which specialises in “building software that meets customer needs”, introduced an extended design thinking framework based on the Double Diamond design process model. [61]

After Zendesk, the main disadvantage of the previous model — design-centricity — was eliminated in the new framework. [16] In the Double Diamond, there is neither a final product at the end of this design mode nor a detailed description of the actual development of that product. Afterwards, once Solving the problem phase starts, the process becomes much more iterative, meaning that designers adapt and change the product plenty of times based on users’ feedback. The Double Diamond model does not reflect this difference in time and effort committed.

At Zendesk, project managers, designers, and engineers took part in testing the new model. That guaranteed that attention was paid equally to every aspect of product creation, from research to development and implementation.

As a result, the **Triple Diamond design process model** emerged. It considers multiple tests and validations with both users and stakeholders.

In the Triple Diamond model, the first two diamonds were not much modified and remained the same as in the Double Diamond model. The design process starts with **Discovery**, where designers first concentrate on research, thus extending the scope of the problem, and then determine the main issues in the **Problem Definition** phase. Once the main problem is articulated, **Solution Discovery** begins. The initial step is ideation: multiple ideas on how to solve the problem are created, assessed, and then the best solutions are combined and transformed into **low-fidelity prototypes**, which are not rich in detail and therefore are not time-consuming. Once those prototypes are finished, they are tested in the **Concept Validation** phase. It is crucial not to skip this step, since if the solution is based on a wrong idea, it is better to detect and fix it before there is much time and effort invested in a flawed concept.

The third diamond is, at its core, an iterative development and validation process, which can and usually will take more time than all other phases combined. That is not only because the later in the development process the designers are, the more time it takes to submit changes to the product, but also because there are many people involved in validation and testing before the launch of the product.

At the beginning of that phase, there is a validated **low-fidelity concept** (lo-fi prototype), which is then developed and after that tested with users. It is also validated with stakeholders if needed. In the Triple Diamond model, this phase is called **Design and Development**. It is then followed by the **Early Access Programme** phase. Tests and validations are performed during both of those phases. As the design process advances, designers face less uncertainty, nonetheless after each test, new problems and alternative solutions emerge.

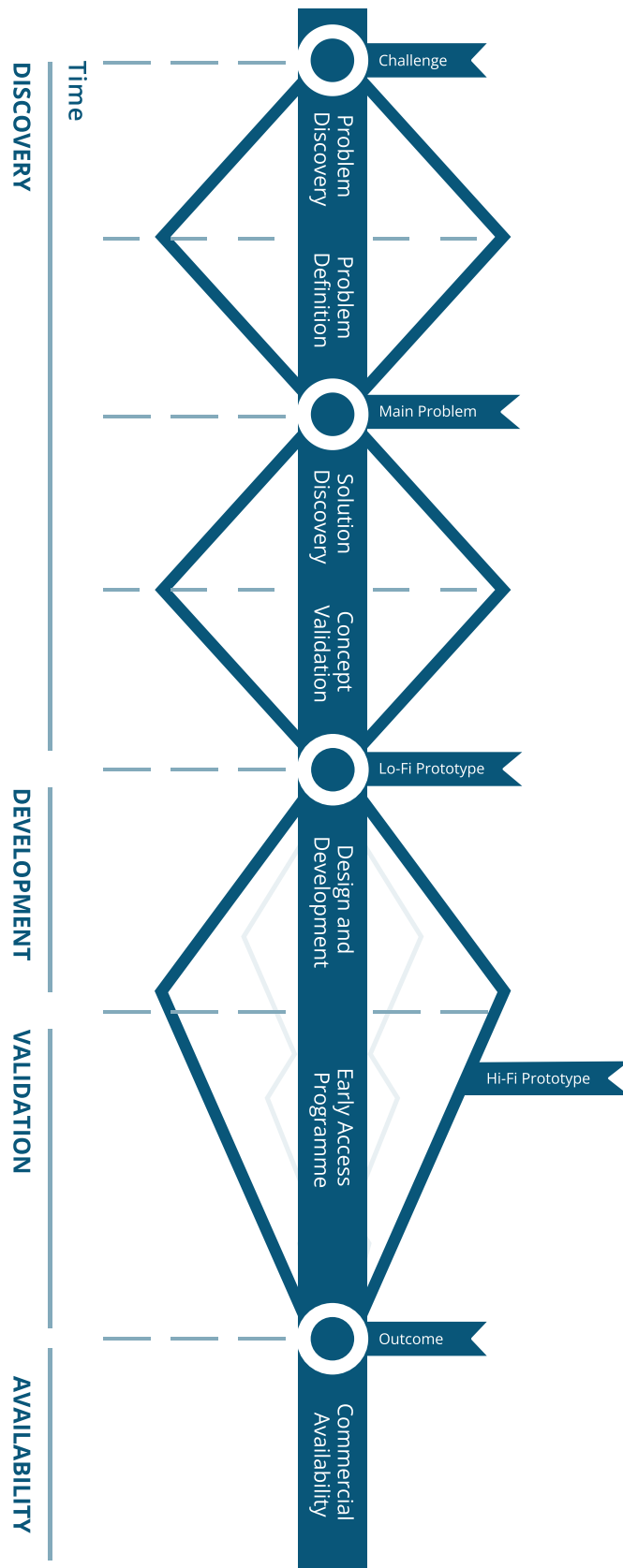


Figure 2.2: The Triple Diamond, Manukian, 2022

The number of tests organised usually depends on the project's scope, budget, deadline, and results of previous tests and validations. For example, even if the project's deadline is five months away but the high-fidelity prototype has proven to be fully functional and has passed the necessary validations, launching the program should be considered.

At the end of the **Early Access Programme** stage, beta versions of the project can be launched and tested on a small number of real users. Then users' behaviours are analysed, and they are asked to provide feedback on the tested project. That concludes this phase of the design process, and once the errors and bugs are fixed, the high-fidelity prototype becomes a finished, thoroughly tested project ready to be launched at full scale.

As Zendesk states, this model may and should be tailored for each unique project. [16] One of the possible alterations is additional research, which can aim at determining how to build a product, or post-launch research in order to assess whether the goal of the project was achieved.

2.4 User testing

The necessity of testing a prototype should not be underestimated. Seeing that designers produce the project and are therefore immersed in it, they may forget the users' problems and frustrations. Only by collecting information on how users interact with the product can designers ensure that everything necessary for the users is included in the product.

As Don Norman says in his book [39], understanding problems “comes about primarily through observation, for people themselves are often unaware of their true needs, even unaware of the difficulties they are encountering.” Thus, if a designer not only receives feedback on the tested product but can also pose such questions that can help determine why the users actually behave the way they do, it can enhance the product. There are different techniques on how to do so, one of which is the **5 Whys method** introduced in 1930 by Sakichi Toyoda, the Japanese industrialist, inventor, and founder of Toyota Industries. [23]

2.4.1 5 Whys method

The goal of this technique is to find the real reason behind a problem by constantly asking *Why?* questions. Let us look at an example.

Problem: students do not attend lectures at the university.

1. “*Why do not students attend lectures?*” The answer to that would be that **they do not have time**. Now, the problem of a time shortage can be solved. In order to deepen the understanding of the issue, it is important to follow up with a *why?* question to expand the understanding of the issue.

2. “*Why do not students have time to attend lectures?*” Let us say that the answer to that question is that **they work during their lectures**. With that information in mind, the work schedule can be considered the problem. If steps are taken to mitigate this problem, though, the core issue would not be solved.
3. “*Why do they work during their lectures?*” Now the answer is: because **they choose to do so**. The next question logically flows from the answer.
4. “*Why do students choose to work during their lectures?*” And the response to that is **because they do not find them useful enough and prefer self-study**. They do not see any particular reason why they should go to the lectures if they can read a textbook written by the same lecturer and then arrange a meeting to ask questions.
5. “*Why do students prefer self-study in the form of reading a textbook?*” And the answer is because **they can learn at their own pace** and review the material at any given time.

So, in this example, the discovered problem evolved **from a busy student’s schedule to a personal preference for a different studying model**. In this kind of activity, more questions may be asked in order to understand the core of the problem. [42]

It is important not only to ask the right questions but also closely watch users’ actions. Usability testing is a type of testing where participants are given a prototype that they must interact with. This type of test is usually recorded and then analysed. [25]

2.4.2 Key Performance Indicators

To keep track of how well users interact with the product, **KPIs** are used. Some of them are:

- **Time on task**: to determine how fast the user follows the flow, time spent on a task is counted. If that time is longer than expected, the user probably does not find the design intuitive and encounters problems understanding what should be done next.
- **User-error rates**: although this indicator tells the number of errors encountered by users, it actually shows how often users followed a different flow than intended.
- **Drop-off rates**: this metric states how often users chose not to do the task because of its complexity. This is a very important indicator because if the drop-off rates are high, users are likely to abandon the product and not perform the desired action at all.
- **System Usability Scale (SUS)**: one of its main advantages is its simplicity and low time consumption. This is a questionnaire that provides 10 questions with 5 possible answers, from “strongly agree” to “strongly disagree.” This scale was created by Kohn Brooke in 1986, and since then it has been used to access a variety of different products and services. Unlike the 5 Whys method, a person can answer the questions without supervision, so more people can take part in the survey. The result of this method is the extent to which the product is usable. This is also its main disadvantage: while it gives precise information about usability, it does not specify the particular issues, and therefore additional tests must be performed. [15]

2.4.3 Subconscious and conscious actions

Users' actions can be well-thought-out and controlled, but once the user does something they are accustomed to doing, they act subconsciously, therefore those actions are fast and automatic. That means that the user may do something even before acknowledging it (e.g., consent to cookies or click on a phishing link seemingly from a bank). That can lead to errors that users may not even notice since they were not engaged in solving a problem. [39]

2.4.4 Biases

According to Collins dictionary [22], "bias is a tendency to prefer one person or thing to another and to favour that person or thing." Below are some biases that can occur in testing:

Confirmation bias (also referred to as yes bias, acquiescence bias, or friendliness bias) is the tendency to overestimate the likelihood that someone will confirm something. The essence of this bias lies in people's willingness to confirm their beliefs on the one hand and their reluctance to consider those that contradict them on the other. In order to avoid this bias in testing, the researcher must collect information using different methods, thus gathering different answers to essentially the same questions. [35]

Social desirability bias addresses the participants' tendency to behave in a way that they believe is socially acceptable, regardless of their actual thoughts and feelings. It can become an issue if research involves collecting data on personal or socially sensitive issues. Possible solutions to that problem include reformulating the same questions and eliminating interviewers as part of the influence on the respondent (non-moderated interviews). [30]

3. Educational Videos

3.1 Typology of educational video

Nowadays, “video is used in teaching in different ways, from lecture capture to peer feedback on video streaming. This is a continually evolving field and defining and mapping the various video forms is an on-going process.” Amongst the advantages of video teaching are time and cost efficiency, enjoyability for students and more time for interactive learning for students. [11]

Hansch et al. provided a list of video production styles. [32] They are classified based on different benefits for users, prompted ways of learning and production methods. An educational video may feature one or multiple production styles listed below:

- **Talking Head:** in the video, the presenter is talking about the topic captured.
- **Presentation Slides with Voice-Over:** video is based on a presentation which is followed by a narration in form of audio.
- **Picture-in-Picture:** a small version of the lecturer’s video can be seen in a corner.
- **Text-Overlay:** a text overlay emphasises the most important information.
- **Khan-Style Tablet Capture:** a visual presentation of the provided information is displayed on a black screen in form of drawn animation captured.
- **Udacity Style Tablet Capture:** in the video, there is a hand writing and drawing on a tablet to illustrate the material.
- **Actual Paper/Whiteboard:** the presented information is written on a paper, blackboard or whiteboard.
- **Screencast:** to demonstrate the material explained, there is a photo or video capture of a screen.
- **Animation:** to present the information, there are animated elements, such as visually presented data, pictures or text.
- **Classroom Lecture:** video contains a captured classroom lecture.
- **Recorded Seminar:** video contains a captured seminar.
- **Interview:** the material is communicated in form of an interview.
- **Conversation:** there are two or more people in the video who are debating a topic.
- **Live Video:** students and a teacher interact with each other in a live meeting or webinar.
- **Webcam Capture:** the presenter’s video is captured with a web camera.
- **Demonstration:** while presenting, the speaker illustrates their speech by sharing their screen.
- **On Location:** the presenter is in a location which may emphasise the topic of the video or act as a background without additional meaning.
- **Green Screen:** the presenter is captured on a green screen which can be changed to any background in post-production.

3.2 Multimedia usage in video

It is also important to highlight the most important information, and in order to do so, Clark and Mayer suggest including both textual and graphically represented information in e-learning. In their research, they identified the importance of providing the explained material in the right combination of audio and visual channels of a video. They also advise against using unnecessary audio since it can distract listeners. [19] It is also stated that segmenting material into sections makes processing information easier. [31]

Since low-pitched and high-pitched voices are perceived differently, the choice of a voice actor is an important one to make. According to a psychological study held in 2016 at the University of Glasgow, UK, both men and women are more likely to view voices with a lower tone as authoritative. [54]

In the study, participants were asked to compare males and females with differently pitched voices and characterise their dominance and trustworthiness.

For both men and women, there was a direct correlation between the pitch and asserted dominance: the lower the pitch, the more often the voice was perceived as dominant. Comparing female and male low voices, though, men's low voices were chosen as dominant significantly more often.

The direct correlation between trustworthiness and the voice pitch was the same, but in this case, females with low voices were chosen as trustworthy slightly more often than males.

Thus, it can be said that males with lower voices are usually perceived as more dominant, whereas females with low-pitched voices are more trustworthy.

It was reported that low female voices are viewed as less dominant and more friendly than men with low-pitched voices. [53] They also discovered that females with high-pitched voices are seen as the friendliest and the most sincere.

Cognitive overload

According to Sweller, an individual can process and understand a certain amount of information at a given time and thus it is important to match the amount of provided material with students' learning capacity. [51]

3.3 Accessibility

3.3.1 A typology of impairments

According to the Americans with Disabilities Act, a disability is “a physical or mental impairment that substantially limits one or more of the major life activities.” [1] Those individuals may encounter various problems interacting with different products and it is important to prevent those issues in order to accommodate them.

There are various types of impairments. Indiana University indicates seven different types of disabilities, as described below. [56]

- **A Visual impairment** is the “inability to see objects, perceive light or color, or correctly judge distances.”
- **An Auditory impairment** is “a partial or total inability to perceive sounds and access audio-based information presented in media.”
- **A Psychological impairment** is “memory loss, reduced attention span, difficulties regulating emotions, and more.”
- **A Neurological impairment** causes “restricted sensory perceptions, mental processes, or motor functions.”
- **A Medical impairment** is “restricted endurance, attention, or mobility, various levels of pain, and fatigue.”
- **A Cognitive impairment** is “loss of memory, reduced attention span, restricted intellectual development, underdeveloped maturity and judgement, or limited problem-solving and logic skills.”
- **A Mobile impairment** is “difficulty with, or inability to use, the hands, feet, arms, or legs.”

According to a list [57] provided by Indiana University, impairments can also be divided into 3 categories depending on circumstances:

- **Permanent** (a person is handicapped for an indefinite amount of time).
- **Temporary** (a person is handicapped for a fixed amount of time).
- **Situational** (a person is not physically handicapped but has the same needs as a handicapped individual).

3.3.2 Accessibility in design

Jesse Hausler, Principal Accessibility Designer at an American software company, ServiceNow [33], states in his article that “accessibility enables people with disabilities to perceive, understand, navigate, interact with, and contribute to the web” and moreover, “only the design itself can cause people with disabilities to have trouble using a product.” Some of the rules of accessible design are the following:

- **Colour** should not be the only visual means of communicating information, meaning that colour should only complement an already distinguishable object.
- Text and background should have a **sufficient contrast** of at least 4.5 to 1. [59]

3.3.3 Accessibility in video design

Additionally, Adobe [52] provides the following advice for accessible videos:

- **Video captions** should be included in videos that have any audio. Video captions may include only spoken speech or an audio description of everything happening in the video.
- **Video transcripts** offering the full video text of everything said in the video should be added as well. This makes videos more accessible for individuals who have reading disabilities and therefore cannot read the captions due to their relatively short display time.
- Also, there may be **audio descriptions** for visual information explaining all important content and providing “clear explanations of what is only happening on screen.”

4. Informational Technology

4.1 VPN

4.1.1 Principle of function

VPN stands for Virtual Private Network, which is a generic name for technologies that enable users to send and receive data across shared or public networks as if their computing devices were directly connected to the same private network.

The advantages of using a VPN include increased security, functionality, and management of that private network. It gives users access to resources that are unavailable in the public network and is therefore frequently used by remote employees or students. [41]

As Cisco states, “businesses often use VPN connections because they’re a more secure way to help employees remotely access private company networks, even when they’re working outside the office.” [17]

4.1.2 Encryption

Although a VPN cannot entirely anonymize web traffic, they successfully improve security and privacy.

The VPN enhances security using the following technologies:

- sent data are encrypted before sending so that a possible attacker would only be able to access the encrypted data and not the raw data;
- the sender is first verified in order to prevent unauthorised users connecting to the VPN,
- to prevent any tampering with the sent messages, message integrity is used. [48]

4.1.3 VPN types

Virtual private networks may be divided into a number of groups depending on the type of connection. Those include Site-to-site VPN, Remote-access VPN, Split tunneling, and Full tunneling VPN. [18]

Site-to-site VPN is used to connect two networks, thus expanding the network between several offices. **Remote access**, or host-to-site VPN, is used to create a secure channel between a corporate network (central office or branch) and a single user connected to another network accesses corporate resources from either a corporate or private computer, laptop, or smartphone. The main advantage of this connection for both users and business stakeholders

is easy, verified access to a company's private resources without being exposed to the open Internet. [48]

Full tunneling VPN means that once connected to a VPN, all internet traffic in all cases goes to a VPN server, thus securing the connection no matter what websites or programs are accessed. On the other hand, **Split tunneling VPN** ensures that only chosen programs have a protected VPN tunnel. This feature reduces traffic overload in data centres and prevents slower connection for the users. It is achieved by reducing the amount of encrypted traffic to absolutely necessary for accessing the needed services, but thus exposing unprotected data to the internet. [38]

4.1.4 Cisco AnyConnect Secure Mobility Client

Cisco Systems, Inc. is a company founded in the USA in 1984 by two Stanford University employees. This company provides cyber security solutions and had 62,000 partners worldwide in 2019. [46]

This company offers different software, including the Cisco AnyConnect Secure Client, which is used to connect to VŠE servers from different computing devices. In order to access a secure VPN connection, users authenticate themselves by logging in using valid credentials or by using a certificate. The second method takes more time since its setup includes visiting the school physically and applying for a digital certificate, whereas in the case of using the Eduroam credentials, there are no other prerequisites in order to access the school VPN. On the other hand, once set up, using a digital certificate is more secure and time-efficient, since once the user is logged in, they do not have to provide a password on that device again. [50]

4.2 NAS

NAS stands for Network Attached Storage. It is directly attached to a local area network, meaning that it connects to an internet router. It is used to share files across devices connected to the same network. That is the reason why the user trying to access NAS has to be connected to the same network as NAS, either physically or virtually via a VPN service.

Thanks to the different protocols it uses, NAS can typically perform all storage functions, including upload, download, and sharing of the data stored. NAS can be therefore used as a foundation for a cloud networking infrastructure where some of the resources and services are available online so that the user does not have to download them. [47]

4.3 Operating systems

4.3.1 Linux

Linux is an open-source operating system first introduced in 1991. Among its advantages are reliability and efficiency. Furthermore, since it is open-source and therefore anyone can freely copy its code on the internet and subsequently modify it, Linux is popular for various systems, such as computers, supercomputers, or cellular telephones. For instance, operating systems made by Google, such as Android and Chrome OS, are based on Linux. Some of the most well-known distributions include Ubuntu, Debian, Kali, Arch, Manjaro, EndeavourOS, Fedora, CentOS Stream, RHEL, AlmaLinux, Rocky Linux, SUSE, Clear Linux, and Gentoo. [12]

GNOME is a free and open-source desktop environment for Linux and other Unix-like operating systems. To simplify, GNOME is the user interface for Linux. Its main features are made to enhance productivity. For example, settings like the VPN connection can be quickly managed in one click in the drop-down menu in the top right of the screen. [28]

To connect to a VPN on Linux in the GNOME environment, several possibilities are available. One of those options is installing the network-manager-openconnect-gnome package. It is done directly in the terminal by running the command “sudo apt install network-manager-openconnect-gnome.” Once the package is installed, the user has to log in using valid credentials or a certificate. [20]

4.3.2 Microsoft Windows

Microsoft Windows (Windows OS, Windows) is a computer operating system that runs on personal computers. As of May 2022, according to Statista, Windows was the most popular operating system, with 76% of the global market. [21]

The first version of Windows was released in 1985, and since then multiple updates have been introduced. In 2022, Windows 10 (2015) and Windows 11 (2021) are the most frequently used versions of system software introduced by the company. [14]

4.3.3 Android

Android is an operating system for smartphones and tablets run by Google Inc. since 2005. According to Statista, Android has held at least 70% of mobile operating systems’ market share worldwide since 2017 up to November 2022. [36] Among the default apps offered by Android is the Google Play Store (called Google Play or Play Store on different devices), which is a library of applications that can be downloaded onto the device. [6]

4.3.4 Apple’s operating systems: macOS and iOS

macOS, as well as iOS, are operating systems introduced by Apple Inc. One of this company’s products’ advantages is connectivity: devices operating both on macOS and iOS may be connected to iCloud, thus letting users share and access their files with all their Apple gadgets. Another of Apple’s unique selling proposition is its good-looking, easy-to-navigate user interface, which is different from the competing operating systems. That in its turn means that to access an application on macOS, a user would need to follow a unique path applicable only to the macOS operating system. [13] In contrast, the iOS system may be compared to the Android operating system since both those systems provide a default application, which is a library of applications that may be downloaded to the device (the App Store in the case of iOS and the Google Play Store on Android). [58] Thus, to download an application available on iOS and Android, users would follow the same path. In 2022, iOS is used on 28% of all mobile devices worldwide. [21]

4.4 Software used for video creation

4.4.1 Pro Tools

Pro Tools, software introduced by Avid Technology, offers plenty of features to make sound recording both for studio-quality voiceovers and music possible. After Avid, Pro Tools provides “the GRAMMY®-winning DAW toolset and plugins”, which are used in the creation of worldwide popular songs. [8]

4.4.2 OBS Studio

As OBS Studio states, it is “free and opensource software for video recording and live streaming”, which is supported by Windows, macOS, and Linux. [49] It offers multiple source capture options, such as whole screen, browsers, or other windows or webcams. OBS Studio offers various output and input settings, like frames per second (FPS) rate, input and output resolution, colour space, and different file formats and encoders.

4.4.3 Adobe Creative Cloud Suite

Adobe Creative Cloud is a project of Adobe Systems, which offers over 25 programs, including those for video editing, motion graphics, and illustrations. [3] One of the many advantages this suite has to offer is dynamic links for importing projects. Changes in that project will be reflected in the dynamic link, sparing time for editing and exporting changed parts of the projects. Another very helpful feature is autosave, which creates multiple copies of the project on its own so that it is possible to access previous versions if needed.

Adobe Premiere Pro

This software provides tools to edit videos, including simple animation and captions. It also offers audio editing, including audio effects that help eliminate clicks and background noises and make sounds more pleasing to hear overall. [5] The main tools for editing audio and video tracks are Selection and Razor tools, along with Rectangle and Type tools for creating shapes and text.

Adobe After Effects

This software is used primarily for motion graphics and visual effects, offering keying, tracking, and animation. [2] One of their main features is layering: separate objects (or sets of objects, even compositions) have their own tracks and are thus animated separately. The position of keyframes, which indicate the state of that layer, can be easily changed, which makes editing uncomplicated. To create the same animation for different tracks, several variants are possible. Firstly, keyframes can be copied and pasted at the same keyframe across multiple tracks at the same time. If the property is not animated on some of the chosen tracks, animation is nonetheless added. The second option is creating a composition consisting of the needed tracks and animating the newly created composition. The third possibility is linking all those tracks to a null object (parent) and adding keyframes to it. All those options are functional, and their application depends on the project and personal preferences.

Adobe Illustrator

Adobe Illustrator is mainly used for creating and editing vector graphics. PC Magazine stated that “Illustrator is the best vector-graphics editing program around.” [55] Amongst its advantages are abundant tools for creating vector graphics, simple content alignment, and various export options. Also, there are different view possibilities, including outline view, which specifically helps work with white elements. [4]

5. Development of technical video tutorials using the Triple Diamond design process model

In the practical part of this thesis, a manual for VŠE students will be created. To do so, the Triple Diamond design process model will be used as a framework for project development.

According to the Triple Diamond design process, the project will be divided into six major steps:

1. Problem discovery.
2. Problem definition.
3. Solution discovery.
4. Concept validation.
5. Design and Development.
6. Final product validation.

During the project, first, **desk research** will be conducted to determine what manual should be made in regard to the discussed topic. Also, a **primary quantitative research** will be conducted to determine whether the found topic was worthy of realisation and what media should be used to communicate with users. Then, to go deeper in understanding the users' problems, **qualitative research** will be held in order to understand what problems and frustrations the users have so that they could be included in the manual.

Next, a solution will be discovered, and a **low-fidelity concept** will be developed. It will be then validated by specialists to ensure its quality and tested by users.

Once the concept is validated and all the necessary improvements are made, it will be fully designed and then developed, resulting in a **Final Product**. The sixth step in the Triple Diamond process is named the Early access programme. In this thesis, usability testing is the priority in this step and therefore it will be referred to as the **Final product validation**. At that point, its usability will be tested, after which the project success will be evaluated. At this stage, the product will be available for launch.

Finally, possible **distribution channels** and the next steps will be discussed. The practical part will be concluded with information about **Timing and Budget** of the project.

6. Problem Discovery

6.1 Target audience and their needs

The initial target audience of the video tutorial was VŠE students who needed clarification of a particular topic related to their studies at VŠE. After having gathered data about students' problems in studies, a possible topic emerged. The students seemed to lack information on how to connect to VPN and to NAS used at the university.

6.2 Quantitative research

To verify the assumption that it was a topic for tutorials worthy of realisation, primary quantitative research was conducted in the form of a Google questionnaire. Its main goal was to determine whether the students would welcome a video tutorial on that topic. To ensure that all the respondents belonged to the initial target audience, the first question determined whether they were VŠE students, which helped eliminate irrelevant responses from the results. The questionnaire gathered 106 responses from the target audience.

VPN is used throughout the faculties, and any student could need it. In contrast, NAS is used only by the KME, which is why the questionnaire was divided into two parts: VPN and NAS. Only those students who had used the service had the possibility to answer questions, which had to determine:

- whether it was easy to access the service,
- whether the respondents would prefer to have a video tutorial on how to access the service,
- how long such a tutorial should be.

6.2.1 VPN

68 respondents had used the VPN offered by the university. One quarter (25.5%) considered connecting to the VPN used by the university very easy (1 on a scale of 4, where 1 is very easy and 4 is very hard). Almost one half (47.2%) chose 2, meaning that they thought that connecting to the VPN was rather easy. Approximately one quarter (27.3%) did not find connecting to these services easy; 7.5% of them chose 4 (very hard).



Table 6.1: Complexity of accessing the school VPN

Generally, students did not find connecting to the VPN challenging, nonetheless, 81.2% of students who had used the VPN before stated that they would welcome a video tutorial on how to connect to the VPN.



Table 6.2: Video tutorial VPN

6.2.2 NAS

As for NAS, 57 respondents had had experience with that service. More than half of them (57.9%) considered connecting to NAS challenging, with 26.3% claiming it to be very hard. Only 17.5% said it was very easy, leaving 75.4% of the respondents who encountered any problems while connecting to NAS.



Table 6.3: Complexity of accessing the school NAS

The majority (80.7%) stated that they would welcome a video tutorial.



Table 6.4: Video tutorial NAS

6.2.3 Length of videos

In both categories, most of the respondents showed interest in tutorial videos shorter than 5 minutes (71% and 74.5% for VPN and NAS tutorials). Therefore, a video tutorial should have been no longer than 5 minutes.

Detailed results of the quantitative research can be found in the Appendix.

6.2.4 Computer literacy

The respondents were also asked to assess their computer literacy. Approximately one half of the students (47%) thought to be rather well-versed with computers (2 on a scale of 4, where 1 is very computer-literate and 4 is not computer-literate at all). 25.5% considered themselves very computer literate, while 27.3% stated that they are not well-versed with computers (3 or 4).

7. Problem Definition

7.1 Qualitative research

Once it was clear that there was high demand for VPN and NAS video tutorials, qualitative research in the form of in-depth interviews was conducted. To cover both topics, apart from being VŠE students, the respondents must have used both VPN and NAS at least once during their studies. Apart from that, it was important to gather people who used different devices and operating systems, such as Windows and macOS, so that they could share their opinions on connecting to those services from different platforms.

The goal of those interviews was to determine what problems they encountered while connecting to those services. Apart from that, it was necessary to discover if they had all the necessary information about those services and why they were used in workplaces.

There were 6 respondents in the interviews. At the beginning of an interview, they were asked to assess their knowledge of technologies. All of them assessed it as rather good (2 on a scale of 4).

Then they were asked to describe how the VPN used by the university worked and why it was needed. 2 respondents out of 6 had no problem explaining it, whereas 4 students admitted that they used those services without thinking about how it worked.

While discussing VPN, students posed several questions:

- I have already purchased a VPN service for personal use, could it be used instead of Cisco AnyConnect?
- Why does Cisco AnyConnect work differently than a VPN that can be freely downloaded from the Internet?
- Do I have to use a VPN if I live in a student's dormitory and use the Eduroam network?
- What would happen if I used Eduroam somewhere in a public space, would I need to connect to the VŠE VPN?
- Can I use Cisco AnyConnect just like any other VPN to protect my data?
- I always forget to add @vse.cz while entering my username, is it necessary?

While connecting to the VPN, all respondents used the university's web page. Some of them stated that the instructions were not very clear as there were many platforms in one article. Furthermore, there were different ways to sign in, which seemed like different steps of signing in. Apart from that, they said that they would like a video tutorial with a voiceover because it would be easier to use than an article.

When asked, how much they would like to know about VPN, the majority (5 out of 6) stated that they would like to know only the necessary information to use it.

Surprisingly, the students showed more understanding of how NAS worked compared to VPN. Half of the respondents seemed to be rather knowledgeable of it, the other half had a general idea about how it worked.

While connecting to NAS they used an instruction provided by KME, nonetheless, they did not describe connecting to NAS as a simple process. Windows 11 users stated that the instructions for Windows corresponded only with Windows 10 and earlier, so they had to specifically search for the same button in their system, which was placed differently in the new layout.

The respondents said that they would welcome a video tutorial on connecting to NAS because it seemed to be a more efficient way to learn how to connect to NAS than reading an article since they could follow the instructions given simultaneously without changing windows to check what the next step was.

The last series of questions for the respondents was what kind of videos they would like if they preferred that way of communication. All of them stated that they would like the video manual to be as short as possible but easy to follow. They also appreciated the idea to separate practical tutorials and general information about VPN or NAS as they thought they would need them in different circumstances. All the respondents stated that they would not like the informative videos to be too official because they thought that they could easily turn out as boring lectures, and they would just skip them and use only the practical tutorials.

7.2 Target audience specification

The original problem was discovered for all VŠE students, but once it was specified, different target audiences emerged.

7.2.1 VPN

Initially, the primary target audience for manuals covering VPN was all students studying at VŠE, both Czech and international, including Erasmus students, regardless of their gender, income, habits, or personality traits. The only constraint was their age: from 18 to 30. Nevertheless, after research, it was clear that there was a secondary target group: VŠE teachers who might need VPN to use the school services remotely.

7.2.2 NAS

Similarly, the initial target audience for manuals covering NAS had been all KME students. The new secondary target audience was teachers at KME, who might need to access NAS to access students' assignments.

People in need of NAS generally did not have any visual or hearing impairments because of the specific courses taught at the department. To study them, students must not have had severe vision, hearing, cognitive, or speech impairments. Most of those courses were taught in the Czech language, so the target audience was expected to understand this language at least on a basic level.

7.2.3 User personas

To make those target audiences more perceivable, two user personas were created for both primary target audience segments. They reflected key features of the respective target group and portrayed them in a more relatable way. While designing solutions, it was important to keep them in mind and make manuals to accommodate those people.

Emma is a 25-year-old student who studies accounting at VŠE. She needs to access school services from different places, including her hometown in Germany, where she comes for holidays. She is a frequent Windows 11 and Android user. Her means of accessing information are primarily Google and YouTube. She does not know how to access the school VPN since she has never used one before. Once she visits the school’s website for instructions, she notices that there is nothing about Windows 11. She feels frustrated about accessing the VPN.

Tom is studying Multimedia in the Economic Practice at VŠE. He is 21 years old, lives in Prague, and studies there. He is a frequent macOS and iOS user. Just a week ago, he started his second school term and needs to use NAS to upload his assignments. That should be done on macOS, and in certain subjects, he has to use Linux which he finds particularly challenging. If he does not succeed in accessing NAS from those devices, he might fail a course. To access NAS, he sometimes needs to access VPN, but he does not know how it is done and feels it is very complicated because there are multiple steps in the process. He also does not know why he would need a VPN to access NAS. For those reasons, he feels frustrated about accessing them.

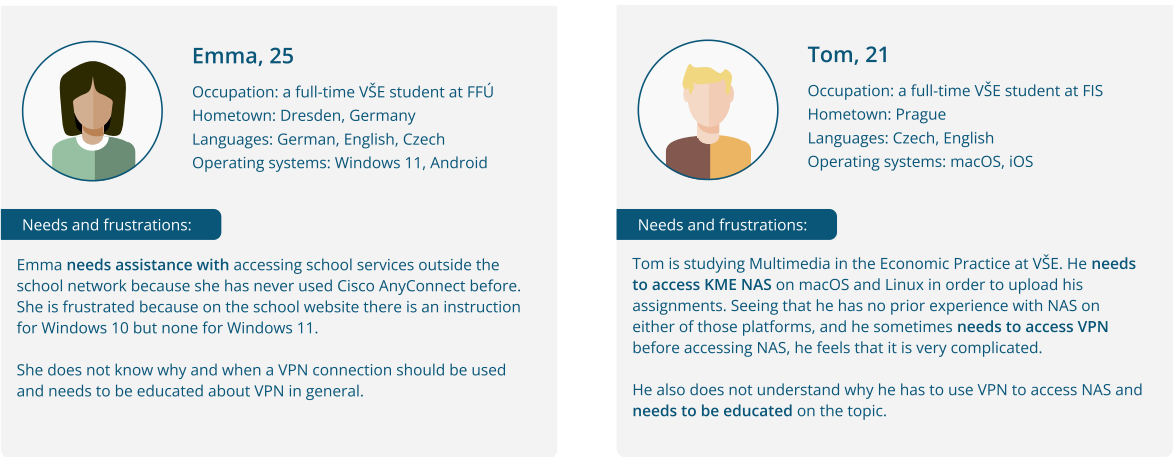


Figure 7.1: User Personas, Manukian, 2022

7.2.4 Problem statements

Problem statements are created for each user persona, reflecting target audiences' needs, frustrations and problems. Those problems are later used for ideation to ensure that the solution is created to successfully solve the most important users' problems. After research, the following problem statements were created:

Emma: My name is Emma. As a student studying at VŠE, who needs to access school services, I am trying to connect to the school VPN. But I cannot understand how to do it from my laptop which operates on Windows 11 because there is no information about this software, which makes me feel frustrated.

Tom: My name is Tom. I am a Multimedia in the Economic Practice student who needs to upload assignments, I am trying to connect to NAS. But it seems very complicated because I do not know the prerequisites to do so, and the instructions seem chaotic, which makes me feel frustrated.

8. Solution Discovery

8.1 Challenges

Before ideation, it was important to define challenges which should have been considered while creating solutions to the users' problems.

1. **Address different target groups.**

It was found that there were two intersecting target groups: the target group that needed NAS was part of a larger group that needed VPN.

2. **Consider different software.**

One of the challenges was not only the different software used by the students but also software updates. For instance, in qualitative research, it was found that users had problems connecting to NAS from Windows 11 since it had a different layout. Further software layout changes could not be predicted.

3. **Create engaging theoretical material.**

Several respondents stated that they would not like for the tutorials to be too official and theoretical, as they would skip them if they found them boring.

4. **Answer the most common questions.**

Most respondents posed similar questions concerning VPN functions, which were:

- How does a VPN for personal use differ from Cisco AnyConnect?
- Could Cisco AnyConnect protect users' data and help access content not accessible in their current location?
- Where is a VPN connection not necessary to access school services?

5. **Economise users' time.**

It was important not to waste users' time, especially since in the research, respondents clearly stated that they would not like to spend much time watching the manuals.

6. **Provide precise information.**

It was also important to provide very precise and accurate information in manuals in order to minimise errors.

7. **Provide easy access to the needed links.**

In order to access VPN and NAS, users were going to need links, some of which were hard to remember.

8.2 Ideation

8.2.1 How might we? questions

In order to address the stated challenges, the HMW technique was used. HMW questions are “seeds for ideation.” [45] They can be made using different techniques, among the most useful for this study were:

- **Questioning the assumption:** HMW completely eliminate wasting users’ time?
- **Going after adjectives:** HMW make theoretical material exciting? HMW address different target groups while keeping the overall communication style consistent?
- **Amp up the good:** HMW create a manual that would be the same for different software versions?
- **Remove the bad:** HMW eliminate the need to remember links needed to access NAS and VPN?
- **Change a status quo:** HMW make accessing VPN and NAS seem easy?

8.2.2 Outcome: answers to How might we? questions

The following are the most favourable ideas for solving the challenges described above:

1. To economise users’ time, disclaimers preventing common mistakes might be used.
2. To educate users, explanations about what VPN and NAS is should be created.
3. To economise users’ time, explanations should be placed separately from the manuals.
4. To make theoretical material exciting, animation could be used.
5. To address different target groups, a consistent neutral tone of voice should be selected.
6. To make videos engaging, a friendly tone of voice should be chosen.
7. To create a manual that is valid despite software updates, one manual for both current and previous versions should be created.
8. To provide precise information about how to access the services, a screencast should be used.
9. To help users understand complex ideas, icons and texts should be used.
10. To let users follow the tutorial in real-time, a video tutorial with a voiceover can be created.
11. To make the videos more engaging and friendly, upbeat background music should be used.

12. To economise users' time, a video tutorial should be as concise and short as possible while communicating all the necessary information.
13. To provide students with the needed links, they should be placed below the manual.
14. To help the users access the video on a mobile device, a QR code might be generated in the description of the video tutorials.
15. To make accessing services seem easy, proper language should be used: no complicated words or sentences.
16. Links to the other manuals should open in other tabs to let the users easily go back.
17. Complicated words and explanations should be avoided in tutorials so that the users are not confused.

8.3 Concept

As a result, it was decided to create **videos in the form of screencast and animation with a voiceover**.

Considering the number of questions asked by respondents during the qualitative research and the overall lack of understanding of how VPN worked, it was decided to separate the videos about VPN into 6 videos:

1. An explainer video about how the school VPN worked with a brief overview of VPN technologies.
2. 5 video tutorials which would be practical tutorials on how to connect to the school VPN from different platforms (Windows, macOS, Linux, iOS, Android).

For the same reasons and in order to keep the video tutorials consistent, it was sensible to make 4 videos about NAS:

1. An explainer video about what a NAS was, covering its specifics at VŠE KME.
2. 3 video tutorials on how to connect to the KME NAS from different platforms (Windows, macOS and Linux).

In total, that would make a set of 10 videos, 2 of which would be explainer videos about VPN and NAS and 8 tutorials about how to access the form different devices. Instructions would be given in form of a voiceover. Additionally, upbeat background music would be used.

8.4 Video types

Since there were multiple topics for the videos made for different target audiences with different production methods, the videos were divided into 4 categories.

- VPN (E): an explainer video about VPN.
- VPN (T): a tutorial about how to connect to a VPN from different platforms.
- NAS (E): an explainer video about NAS.
- NAS (T): a tutorial about how to connect to a NAS from different platforms.

Video Types	Explainer	Tutorial
VPN	VPN (E)	VPN (T)
NAS	NAS (E)	NAS (T)

Table 8.1: Video types

8.4.1 Explainer videos

The main goal of explainer videos was to introduce the topic and inform about some important notions concerning it. Due to that reason, those videos should not have been too long (3-5 minutes). The information should have been structured and rather entertaining so that it would be well-perceived. To illustrate this information, an animated video was created in the form of suitable icons and important notions: VPN, Remote Access VPN, and Split Tunnelling. It was decided to create those videos as an animation with a voiceover, also using text overlay.

8.4.2 Video tutorials

Video tutorials should have been as concise as possible, being no longer than 3 minutes. Their primary aim was to guide users through connecting to a service and give answers to possible questions that could occur while accessing those services. It was decided to produce those videos in form of screencasts with a voiceover. Text overlay was used to prevent mistakes and emphasise the information which might be useful for the users.

8.5 Key Performance Indicators

KPIs are usually used to determine whether the project was successful.

It is important to choose applicable KPIs for the project and then state expected values so that it is possible to determine how well the project was realised by comparing the desired outcome and the achieved results.

8.5.1 KPIs for explainer videos

To determine whether explainer videos were understandable and useful to the users, 3 KPIs were chosen:

1. Drop-off rates: 40%, meaning that only 2 out of 5 users would skip explainer videos knowing that they would be asked questions on the topic,
2. Test score: 60% correct responses, meaning that an average user would answer correctly at least to 3 out of 5 questions regarding the topic of the watched video.
3. Usability scale: an average user would at least agree with positive statements about the usability of the videos, e.g., “I agree that the information presented in the video was understandable.”

8.5.2 KPIs for video tutorials

As for video tutorials, it was crucial to measure how well the users could follow the instructions in the videos, therefore following KPIs were chosen:

1. Time on task: up to 7 minutes on each task (including Cisco AnyConnect downloading), if connecting to VPN, and up to 6 minutes connecting to NAS, while using video tutorials.
2. User error rates: 20%, meaning that no more than 1 out of 5 users would encounter problems while accessing the service presented in the video tutorial.

8.6 Customer Journey

Once the video types and production methods were selected, it was crucial to create Customer Journeys. A Customer Journey depicts a happy path of a user from the discovery of a product to its promotion, all while concentrating on their needs, feelings and actions.

The Customer Journey created in this project was created by combining and modifying two frameworks: STDC and AIDA.

8.6.1 Customer Journey for VPN videos

Customer Journey 1 was created for users who only needed to access VPN. This target audience was presented by the user persona called Emma.

Customer Journey 1	Needs and Pains	Actions	Feelings
Discovery	User needs to access VŠE services remotely but does not know how to do that	User looks for a manual how to do that and sees that they need VPN	User is frustrated and worried because setting up VPN seems too hard
Awareness	User needs to access VPN	User searches for a manual how to access VPN and sees a video tutorial	User is afraid that they might not be able to access VPN without assistance
Consideration	User needs a concise video tutorial	User looks at the video tutorial and sees that it is short	User is hoping that a video tutorial will help in accessing VPN
Use	User needs to use right credentials to access VPN	User follows the video in the background or on a small screen. If needed, they pause the video	User is content and relaxed once they have their problem solved
Care	User is happy accessing VPN was easy and is now curious how VPN works	User opens link leading to educational video about VPN	User is satisfied that they have a have a way to find out more about the topic
Promotion	User needs to advice friends a good way to connect to VPN	User gladly promotes video tutorial they watched	User is happy to help their classmates

Table 8.2: Customer Journey 1

Key takeaways from Customer Journey 1 were the following:

- users do not want to waste their time and immediately open the tutorial,
- once the users solve their problem, they consider watching a VPN (E) video,
- users mostly watch tutorials either in the background or on small screens,
- users pause the video when they need more time to follow instructions,
- it is important to highlight what credentials need to be used (InSIS or Eduroam).

8.6.2 Customer Journey for NAS videos

Customer Journey 2 was created for those users who needed to access a school NAS. This target audience was presented by the user persona called Tom. In order to access the NAS outside of the school network, users had to connect to the school VPN, thus making their journey more complicated. That is the reason why Customer Journey 2 includes steps from Customer Journey 1.

Customer Journey 2	Needs and Pains	Actions	Feelings
Discovery/NAS	User needs to hand in their assignments remotely but does not know how to do that	User looks for a manual how to do that and sees that they need NAS	User is frustrated and worried because accessing NAS seems hard
Awareness	User needs to access NAS	User searches for a manual how to access NAS and sees a video tutorial	User is afraid that they might not be able to access NAS without assistance
Consideration	User needs a concise video tutorial	User looks at the video tutorial and sees that it is short	User is hoping that a video tutorial will help in accessing NAS
Discovery and Awareness/VPN	User finds out that they need VPN connection to access NAS	User finds a link to video tutorial how to access VPN	User is afraid that they might not be able to access VPN without assistance
Consideration/VPN	User needs a concise video tutorial	User looks at the video tutorial and sees that it is short	User is hoping that a video tutorial will help in accessing VPN
Use/VPN	User needs to use right credentials to access VPN	User follows the video in the background or on a small screen. If needed, they pause the video	User is content and opens previous video to connect to NAS
Use/NAS	User needs to use right credentials to access NAS	User follows the video in the background or on a small screen. If needed, they pause the video	User is content and relaxed once they have their problem solved
Care	User is happy accessing VPN and NAS was easy and is now curious how VPN and NAS works	User opens link leading to educational video about VPN and NAS	User is satisfied that they have a have a way to find out more about the topic
Promotion	User needs to advice friends a good way to connect to VPN and NAS	User gladly promotes video tutorials they watched	User is happy to help their classmates

Table 8.3: Customer Journey 2

Key takeaways from Customer Journey 2 were:

- users have to watch two tutorials in order to access KME NAS: VPN (T) and NAS (T),
- users are more motivated to watch multiple tutorials if they are not too long,
- while accessing VPN and NAS, users need different sets of credentials (Eduroam credentials for VPN and InSIS for NAS).

8.7 Edge cases

Despite thorough Customer Journeys, sometimes users do not follow happy paths, due to unusual circumstances, needs, or different backgrounds. Those users should be considered explicitly in the development phase.

8.7.1 International students

VŠE hosts international students every year, and those students might need to access VPN at any moment. It should be remembered that they might not speak Czech. [40]

8.7.2 Students already connected to the school network

Some students might be already connected to the school network and be unaware that they do not need VPN to access needed services. Therefore, to economise their time, there should be a disclaimer that they should not watch the tutorial to access the service needed.

8.7.3 Students with impairments

Some students who need to access the school VPN may have vision and hearing impairments. Videos should be accessible to those students as well.

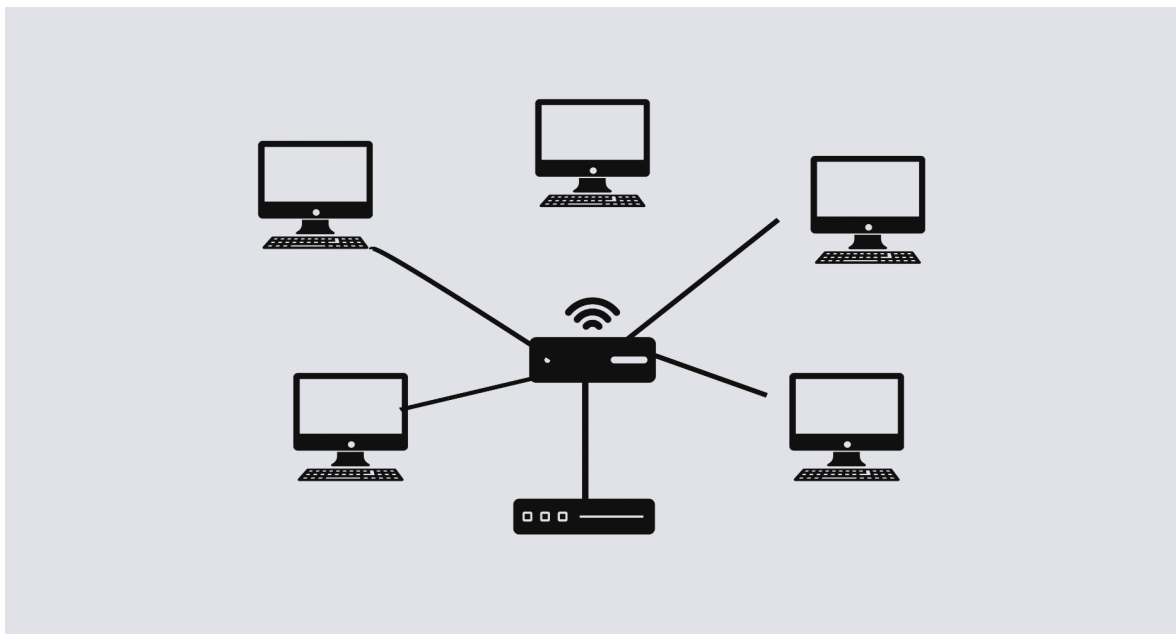
8.8 Informational structure and consistency in videos

For the videos to be understandable, it was important to use appropriate simple language, structure information coherently, not overwhelm users with too much information, and leave out irrelevant details. At the same time, it was crucial to mention everything that could be challenging for the users and help them successfully access VPN and NAS and educate them about those topics.

It was also important to keep the videos consistent, not only for design reasons but to increase efficiency in the production phase, namely voiceover recording so that it could be recorded once and used in different videos in postproduction. That is the reason why identical texts were written wherever possible.

9. Concept Validation

Concept Validation was crucial to determine whether the concept would be considered advantageous for users. In order to test the concept, low-fidelity videos were made (screen recording with voiceover and low-fidelity animation). Apart from the usability testing with respondents, the concept was also validated with experts on the topic discussed in the videos so that there would be no technical mistakes. All the given feedback was processed and reflected in the product.



Username	AD\mark04
Password	*****

Figure 9.1: Low-fidelity prototypes, Manukian

9.1 Usability testing

The goal of this testing was to determine how users would interact with texts, audio, and visuals used in the videos; collect feedback, and then improve the prototype. The test was performed in the form of a lab-moderated test.

As there were 10 videos in total but some of them were similar in terms of actions users had to perform, to avoid inaccurate results, one respondent could test only one set of videos, making it 2 different paths for different respondents, which can be seen in the chart below.

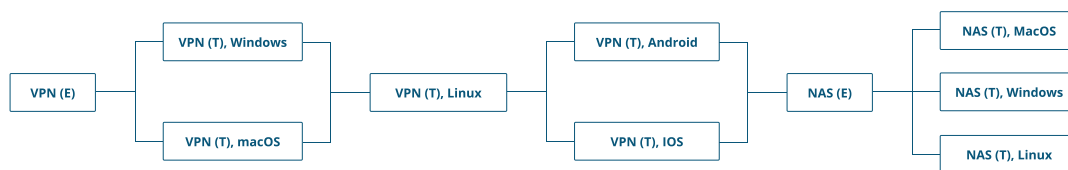


Table 9.1: Concept test

Also, an **A/B test** was conducted to determine which voiceover was better perceived. There were two versions of voiceovers, performed by a male and a female voice, which were distributed in such a way that every participant heard both voices and expressed their opinion on them.

In total, six VŠE students, aged 18–30 and speaking Czech, took part in the concept validation. At least 2 of them were active MacBook and iPhone users, 2 of them were active Android and Windows users, and 1 user had used Linux previously.

A full script of the testing can be found in the Appendix. The participants were asked questions to determine the following:

- Does a voice pitch influence the user's perception of the content?
- Is there any missing information in the videos?
- Is there any unnecessary information in the videos?
- How do users feel about the icons?

9.2 Insights

All the participants expressed that they would use videos as tutorials that they would watch while connecting to VPN and NAS. Although, 2 participants said that they may have skipped long videos about NAS and VPN since they do not need to understand how they work to use them.

Key insights were the following:

- The male voiceover was preferred slightly more than the female one.
- Minimalist animations did not disturb viewers from the narrative.
- At the end of the VPN (T) Windows video, there were too many details which were not needed.
- Text should have been consistent and neutral, not too official, which it was not.
- It would be useful to highlight all the keyboard shortcuts used in the video and have them written on the screen or below the video.
- It would be useful not only to give the links but also to spell them while typing.
- Photos did not look consistent in the videos, only icons and text should have been used.

9.3 Biases

Seeing that the study was performed in a friendly student environment, the following biases might have taken place.

1. **Confirmation bias:** this bias expresses that people tend to do things the way they are used to.
2. **Social desirability bias:** this bias describes the tendency to respond to inquiries in a manner that will be appreciated by others.

Due to that reason, it is important to keep in mind that more students might have skipped the educational videos more often than it was determined in the testing.

10. Design and Development

The main advantage of designing and developing products at the same time was that all the necessary redesigns may have been made even after the development has begun. Generally, the Design and Development phase implies that preproduction, production and postproduction may happen simultaneously. In this project, the production and postproduction phases were also intertwined. Thanks to that, everything that could be produced at a certain time, was indeed produced and therefore it was possible to be flexible in that phase of the project.

In order to keep the structure clear, this chapter will not reflect the order in which each part of preproduction, production and postproduction took place.

10.1 Preproduction

First, a voiceover actor was chosen, and then, a Contract of the Creation and Use of Artistic Performance Licensing Agreement was prepared for signing. After that, a recording studio was reserved and the exact devices on which the recording would be made were determined.

10.1.1 Scripts polishing and formatting

To perfect the scripts, insights formulated after the concept validation were used to create the final version of the text, which can be found in the Appendix. To make recording efficient and easy, it was decided to divide the texts into sections and number them so that editing went without additional complications. The changes in the texts were made according to the actor's preferences, including adding extra spaces and additional exclamation marks to bring the actor's attention to complicated words. The hardest phrase was expected to be “připojit při přihlášení” as there were many “ř” sounds there. Since it was a phrase necessary for accessing the school VPN, it could not be avoided.

10.1.2 Necessary inconsistencies in the video tutorials

Even though the videos were created to be almost identical, there were instances when it was sensible to make some major changes in the videos to benefit the users. One example is the tutorial video on accessing NAS on macOS: that is the only software which allows the users to add multiple folders from one server at a time and therefore for that video, the voiceover text was changed to reflect those differences. The server link used in the video differed as well.

10.2 Production

10.2.1 Screen capture

To capture screens on different operating systems, different software was used.

OBS Studio

This software was used to capture screens on Windows, macOS, and Linux platforms. While recording, the resolution was set to at least High Definition (HD) 1280×720 (for Windows) and to Ultra High Definition (UHD) 3840×2160 (for macOS and Linux). For all three, the 60 FPS rate was set.

Default screen recording software for iOS

On an iOS device, a built-in program for screen capture was used. The properties of the output video were 33.121 FPS, 828×1792 .

Default screen recording software for Android

On an Android device, similarly, a built-in program for screen capture was used. The properties of the output video were 42.819 FPS, 1080×2400 .

10.2.2 Sound capture

Contract with the voiceover actor

A contract of the Creation and Use of Artistic Performance Licensing Agreement had been signed before the sound recording. The text of that contract can be found in the Appendix.

Microphone

To create a good sound, a Rode NT2-A condenser microphone was used. As Rode [44] states, it offers “pristine audio quality and flexibility in equal measure.” To avoid additional delay in sound, soundproof panels were situated behind the actor.

Pro Tools

This software was used to record the voiceover. Minor manual edits were made using Pro Tools as well.

10.2.3 Challenges

During the sound recording, some challenges emerged.

Firstly, the actor was sick, and although his performance did not deteriorate in the slightest, the recording had to be held over two days, two weeks apart from each other.

Secondly, in the text, there were quite a lot of complicated (sometimes English) words. Those were, for instance, Eduroam, Cisco AnyConnect, and GNOME. Those were edited in the text as Eduróm, Cisko Enykonnekt and Noum. Even though, sometimes more than 10 takes were needed to record one sentence. The hardest phrase to pronounce was “Poté povolte načtení systémového softwaru z aplikace Cisco AnyConnect kliknutím na tlačítko Povolit.”

10.3 Postproduction

10.3.1 Background music choice

To make videos feel more dynamic, upbeat background music was added. The used song Tutorial was part of the Building Hope album, which was composed by Edmund J. Jr King and published under the Sonoton Virtual label in 2021. [9]

One of the reasons it was chosen was its structure, which let it be edited and mixed to meet the video length requirements. The other advantage was that there were two versions of the same composition, one of them featuring a violin and guitar, which made it possible to combine different versions of that song without additional complications. To keep the videos consistent, this song was the only one used for all 10 videos.

10.3.2 Attributions

There were many different icons used in explainer videos, which were distributed under the Creative Commons Licence. Under this license, icons can be freely used if they are credited. A list of the necessary attributions can be found in the Appendix.

The music used in the videos was produced by Studio Fontána. After its arrangement with VŠE, their compositions may be used in students' works to any extent if those compositions were produced as part of VŠE courses. [10]

10.3.3 Editing in Adobe Premiere Pro

For editing videos, Premiere Pro was chosen. Resolution and FPS rate were set differently for each video according to their original properties. It was important to determine in what order audio and screencasts had to be processed since both were recorded so that they could be flexibly edited.

It was decided that voiceover should have been the basis for the videos. According to the Customer Journeys, users would most likely use the voiceover as guidance and pay less attention to the screencast itself, unless they had a problem and needed to consult the screencast. That is why it was extremely important to have a spirited narrative at a good pace. The voiceover for all 10 videos was edited in Premiere Pro in order to keep the videos as consistent as possible. The effects used for those audios were the following:

- Speed (105%),
- Graphic Equalizer (20 bands),
- Automatic Click Remover,
- DeNoise,
- Treble,
- Surround Reverb.

The next step was adding and editing the screencast so that it followed the audio word for word. That is the reason why video editing took longer than originally expected. Sometimes, some animations had to be added. For instance, Android screencast did not hide character input, and thus the password used to access VPN could be seen on the screen.

10.3.4 Editing in Adobe After Effects

Once audios were edited After Effects projects were created. The resolution was set to UHD 3840 x 2160 and FPS rate to 60.

Audios were imported to After Effects in the form of dynamic links, just as icons were, which were edited in Illustrator. Some of the properties used to animate those icons include the following:

- position,
- opacity,
- scale,
- rotation (y-axis),
- trim paths.

10.3.5 Accessibility

Accessible design can also be described as inclusive design, which can be used regardless of ability, race, economic status, language, age, gender, or other users' characteristics. This design is based on the notion of equity, which means different amounts of opportunity and support depending on the individual, which in turn results in the same opportunities for each person.

One of the most important factors that should be considered while watching any kind of audio-visual content is impairments. Those can be permanent (e.g., permanent loss of vision), temporary (e.g., loss of vision for a fixed period of time, for instance, after a surgery), or situational (e.g., the user cannot see visual content as they are looking at another screen).

According to the Customer Journey described previously, users are highly likely to face situational visual impairments, and thus it is highly important to consider them while producing the videos.

Colours

In order to make the video accessible for people with partial visual impairments like colour blindness, the chosen colour palette was monochrome, with HEX #095679 used as the background and HEX #FFFFFF for the texts and visuals.

Monochrome palettes are colourblind-friendly since colours do not differ in hue and thus cannot be confused.

The used colours' contrast ratio is 8.02:1, which means that both text and graphic components are easy to read for people with up to 80% vision loss. According to accessibility standards, people who have lost more than 80% of their vision usually use assistive technologies like screen readers. [59]

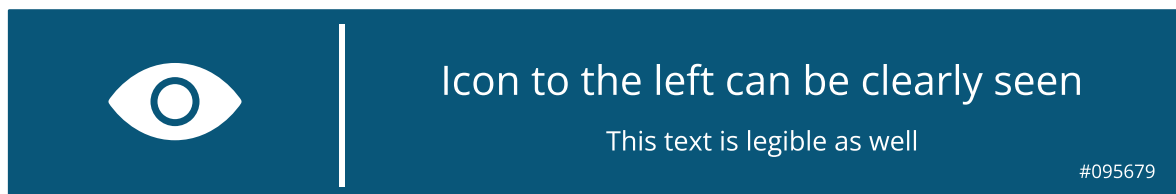


Figure 10.1: Colour palette, Manukian, 2022

To accommodate those individuals, the text of video tutorials is very descriptive, mentioning everything important that one can see on the screen. Words like "here" or unspecific phrases like "turn on the function" were deliberately eliminated to avoid any possible confusion. That also benefits users who follow videos on small screens or in another tab and are thus situationally impaired and cannot see details on the screen.

Dark themes

For people sensitive to bright light, video tutorials feature applications with dark themes, which can encourage students to use them as well, thus reducing electricity consumption.

Font

The chosen font Open Sans is a sans-serif font designed by Steve Matteson and commissioned by Google in 2011. [29] The font is open source, as a result of which everyone can both contribute to it and use it without any restrictions. The font is featured on more than 96 million websites all over the world. That is due not only to its legibility but also to the number of languages it supports, which is 120 only in Europe, among which is the Czech language.



Figure 10.2: Open Sans, Manukian, 2022

Captions

Since people with hearing impairments could encounter problems while using video tutorials and especially while watching explainer videos, subtitles were created. Captions could benefit not only impaired individuals but also international students, for whom subtitles in English were created as well.

Seeing that the target audience of videos regarding NAS was supposed to have little to no impairment, subtitles for them were not sensible and thus were not created.

10.4 Final Product

10.4.1 Key deliverables

As a result, 10 videos concerning VPN and NAS were created, 2 of them being explainer videos, and the other 8 were tutorial videos guiding the users through the process of accessing those services. A link to the produced videos can be found in the Appendix.

Explainer videos: 2 animated videos with a voiceover and background music aiming at educating the watchers about the discussed topic.

1. **VPN (E)** gives an overview of what a VPN is and information on how it is used at VŠE.
2. **NAS (E)** explains what a NAS is and describes its main characteristics. Also, its use at VŠE is discussed.

Video tutorials: screencasts with overlay-texts accompanied by a voiceover and background music. Their primary goal was to help the watcher to access the needed service as easily as possible.

1. **VPN (T), Windows** guides the watcher through downloading Cisco AnyConnect and its setup on Microsoft Windows.
2. **VPN (T), macOS** guides the watcher through downloading Cisco AnyConnect and its setup on macOS.
3. **VPN (T), Linux** guides the watcher through downloading Cisco AnyConnect and its setup on Linux.
4. **VPN (T), iOS** guides the watcher through downloading Cisco Secure Client and its setup on iOS.
5. **VPN (T), Android** guides the watcher through downloading Cisco Secure Client and its setup on Android.
6. **NAS (T), Windows** guides the watcher through accessing the school NAS and adding the right folder on Microsoft Windows.
7. **NAS (T), macOS** guides the watcher through accessing the school NAS and adding the right folder(s) on macOS.
8. **NAS (T), Linux** guides the watcher through accessing the school NAS and adding the right folder on Linux.

10.4.2 Timing

All tutorials were under 3 minutes as originally planned, explainer videos were no longer than 5 minutes as well. In total, animated explainer videos were 7 minutes and 40 seconds long and all screencast tutorials combined lasted 18 minutes and 10 seconds. The longest tutorials were 2 minutes and 40 seconds, the shortest ones were 1 minute and 50 seconds.

That totals in 25 minutes and 50 seconds of video produced. The table below gives information about each video.

Final Product	Type	Length, min:sec	Resolution, pixels	FPS
VPN (E)	Animation	4:30	3840 x 2160	60
NAS (E)	Animation	3:10	3840 x 2160	60
VPN (T), Windows	Screencast	2:10	1920 x 720	60
VPN (T), macOS	Screencast	2:40	3840 x 2160	60
VPN (T), Linux	Screencast	2:40	3840 x 2160	60
VPN (T), Android	Screencast	2:25	1080 x 2400	30
VPN (T), IOS	Screencast	2:15	828 x 1792	30
NAS (T), Windows	Screencast	2:15	1920 x 720	60
NAS (T), macOS	Screencast	1:50	3840 x 2160	60
NAS (T), Linux	Screencast	1:50	3840 x 2160	60
Total	N/A	25:50	N/A	N/A

Table 10.1: Final Product

10.4.3 Export properties

Animated videos were exported in UHD as well as tutorials for macOS and Linux. Tutorial videos for Windows were exported in HD. Tutorial videos for iOS and Android devices were exported in 828 x 1792 and 1080 x 2400 resolutions respectively.

Original FPS rate was used for all operating systems except for iOS and Android which were both set to 30 FPS. All videos were exported in Apple ProRes MXF OP1a.

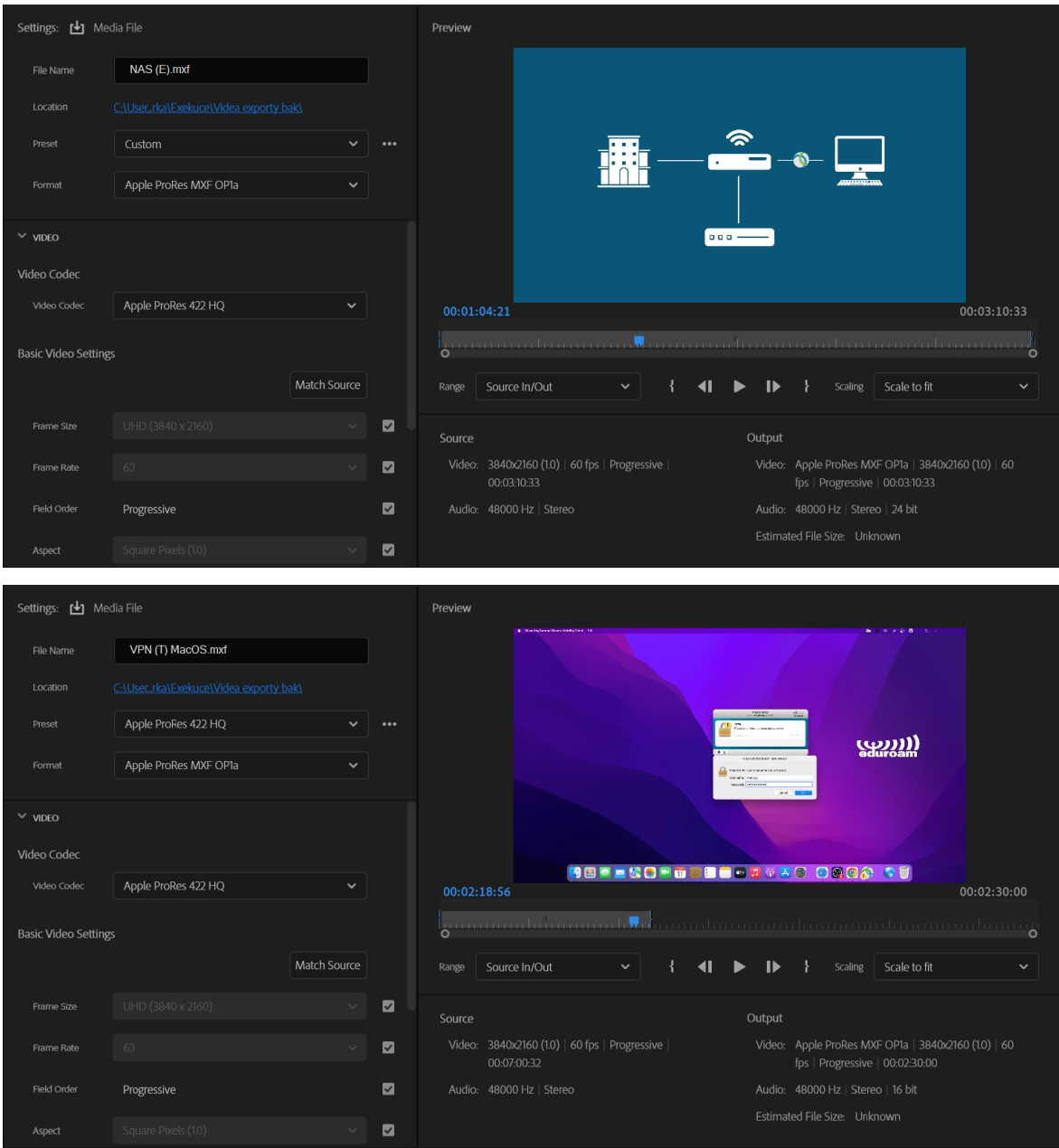


Table 10.2: Export properties

11. Final Product Validation

At the beginning of the project, different KPIs were chosen for different videos.

Their primary goal was to determine if users found the videos useful, would benefit from them and would promote them to their classmates if given the chance. The final product underwent usability testing and was also validated by a specialist on the topic discussed in the videos. All 10 produced videos were tested.



Figure 11.1: High-fidelity prototypes, Manukian, 2022

11.1 Expected KPIs for video tutorials

For video tutorials chosen KPIs were **Time on task** and **User error rates**.

1. **Time on task** indicator was set to **7 and 6 minutes** for the VPN and NAS tutorials respectively, which meant that a user should not have spent more time on accessing those services while using the video tutorials. The time difference between the VPN and NAS tutorials was mainly caused by the time needed to download the Cisco AnyConnect client.
2. **User error rates** should have been **no more than 20%** so that it could be surely stated that no more than 1 out of 5 users would encounter problems while accessing a VPN or NAS.

11.2 Expected KPIs for explainer videos

For Explainer Videos chosen KPIs were a **Test score**, **Drop-off rates** and a customised **SUS**.

1. **Test score** would measure how well the video communicated the information to the users. After having watched the video, the respondents had to answer some questions about the covered topics. **The lowest expected test score was 60%**. In order to compare whether users' knowledge of the topic deepened, the same questions were offered before watching the video.
2. **Drop-off rates** should have been **no more than 40%**, which would mean that most respondents would find the videos informative and engaging enough to watch them. Interpreting results could be complicated, so if the user skipped the video, they were asked to specify why.
3. **SUS** is usually used in an industrial context and therefore can be applied to many respondents. [15] For this testing, it was decided to use a simpler version of the SUS: 10 positive sentences about the videos were created and respondents were asked to express the extent to which they agreed or disagreed with them (5 for strongly agree, 4 for agree, 3 for not sure, 2 for disagree, and 1 for strongly disagree). In order to state that most users found the video useful, understandable, structured, and well-edited, a **minimum of 3.5** had to be reached.

11.3 Usability testing

The primary goal of the testing was to determine whether the project was successful, satisfied users’ needs and solved their problems. To do so, a lab–moderated test was performed during which different KPIs were assessed for different video types. A full script of the testing can be found in the Appendix.

1. To determine VPN (T) and NAS (T) performance, following indicators measured:
 - **time on task**,
 - **user error rates**.
2. To determine VPN (E) and NAS (E) performance, following indicators determined:
 - **drop-off rates**,
 - **SUS**,
 - **test scores**.

Similarly as in Concept validation, to avoid inaccurate results, 2 different paths for different respondents were made. This algorithm differs from the previous one in accordance with Customer Journeys and later testing, where it was clear that the educational videos were mostly watched after the user had accessed the needed service.

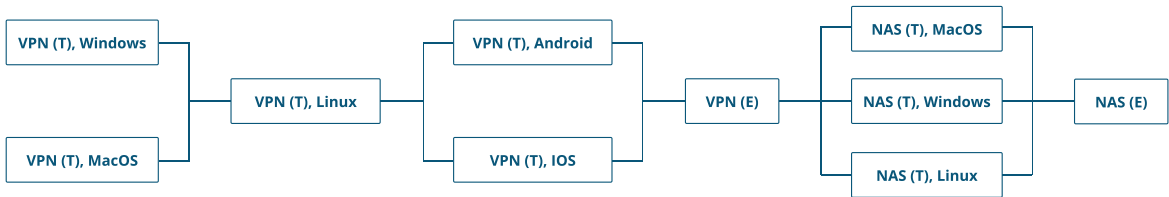


Table 11.1: Validation algorithm

The chosen respondents had to have no previous experience with connecting to VŠE VPN and NAS. Due to that reason, the targeted audience was broadened to students at any university. They had to be aged 18-30 and speak either Czech or English. At least 2 of them were active MacBook and iPhone users, 2 of them were active Android and Windows users, and 1 user had used Linux previously.

There were 6 respondents in total, which means that the Explainer videos were tested by 6 people, whereas the least tested videos were NAS (T) which were tested by two people each.

11.4 Results and Insights

11.4.1 Tutorial videos

After sharing their personal experience with VPN (4 out of 6 respondents had installed it before for personal use), they were asked to connect to the school VPN. The respondents were provided with VPN (E) and VPN (T) videos, a school article on how to set up the school VPN and credentials for Eduroam and InSIS so that it could be assessed if they would choose the right credentials. As predicted in Customer Journeys, initially, **5 out of 6 respondents skipped the Explainer videos**.

Time on task

On average, it took users **5 minutes and 25 seconds** to access VPN which was faster than expected. Linux proved to be the most challenging operating system for users due to their lack of experience with it.

User error rates

During the testing, only **two errors occurred**, both of them were users trying to log in with the wrong credentials. Nonetheless, they realised their mistake without assistance thanks to the overlay text on the videos.

Concept Validation Results	Respondents	Time on task	User error rates, %
VPN (T), Windows	3	5:10	17
VPN (T), macOS	3	5:42	17
VPN (T), Linux	6	6:50	0
VPN (T), Android	3	5:20	0
VPN (T), IOS	3	3:10	0
VPN (T), Average	3	5:25	7
NAS (T), Windows	2	4:47	17
NAS (T), macOS	2	4:15	17
NAS (T), Linux	2	4:10	0
NAS (T), Average	2	4:25	11
Expected	N/A	max. 6/7	max. 20

Table 11.2: Final Product Validation: Explainer videos

11.4.2 Explainer videos

Test score 1

Before watching VPN (E) and NAS (E), the respondents were asked a set of questions to assess their knowledge on the topic covered in the video. Thus their initial knowledge of the topic could be assessed and later compared to the second test score to determine how much new information users apprehended. It was found that understanding of NAS was worse than VPN and on average users could answer correctly on **40% of the posed questions**.

Drop-off rates

One of the respondents skipped VPN (E) video because of sufficient knowledge of the topic and good performance on the first test. None of the respondents chose to skip NAS (E) video, even though they were offered that possibility. That could be due to social desirability bias, which means that when participants knew that they would be assessed on the topics presented in the videos, they were more likely to watch them attentively.

Test score 2

In the second test which took place after the respondents watched the videos, the average test score was **45% better** in comparison with the initial results. One of the reasons why the respondents did the second test so well was that once they had seen the questions they had to answer, they attentively listened to the videos trying to hear the necessary information. According to the respondents, those questions helped them know what the most important information in the videos was and they could concentrate on it.

System Usability Scale

VPN (E) was given a **4.1** average on the customised SUS, where 5 meant strong appreciation of the video, whereas NAS (E) received a **4.4** score. Since NAS (E) was shorter and gave less complicated notions, it was easier to understand and thus users felt more confident about their understanding of the topic in the video.

Concept Validation Results	Respondents	Drop-off rates, %	Test score 1, %	Test score 2, %	SUS, 1-5
VPN (E)	6	17	45	80	4.1
NAS (E)	6	0	35	90	4.4
VPN&NAS (E) , Average	6	8.5	40	85	4.25
Expected	N/A	max. 20	50	min. 60	min. 3.5

Table 11.3: Final Product Validation: Tutorial videos

11.4.3 Feedback

Overall, respondents appreciated the structure of the videos. It was stated that they were not overly complicated and communicated absolutely necessary information for understanding the topic discussed in the videos.

All the respondents preferred video tutorials to written instructions in form of an article on the school website.

The voiceover was slow enough to give the users time to understand what it was actually saying and fast enough to keep a spirited narrative.

Overlay labels for Eduroam and InSIS credentials were highly appreciated and helped users solve problems which occurred during accessing VPN and NAS.

Otherwise, the simplistic, minimalist design of explainer videos was complimented because it did not distract the respondents from the topic but at the same time duplicated the most necessary information so that it received users' attention.

Some of the feedback the videos received was the following:

"I loved that there was a voice which guided me through the process. I liked the sound of the voice as well."

"I found the video about VPN quite useful and would rewatch it if I forget something."

"That is the same VPN and NAS that I use at work, and now I finally know why it is used!"

"I liked the animations, they kept the videos fun."

"I wish I had such videos on every school service."

"I believe that after watching this video I will be able to access VPN and NAS without additional assistance."

"I appreciate that there are subtitles in the videos, I think many people could use them."

"I think that videos are much better at explaining such topics. And there could be situations when I wouldn't want to read or listen to instructions, then I could simply open the video without sound and follow the video."

11.5 Project success evaluation

In the usability testing, multiple KPIs were used to measure whether the videos satisfied the users' needs and solved their problems. Those KPIs were:

1. **Time on task.** While connecting to NAS and VPN while following the corresponding videos the average time on task was approximately **10% less than originally planned**.
2. **User error rates.** The participants encountered **fewer errors than expected** as well. Moreover, to solve those problems, the videos were sufficient.
3. **Drop-off rates.** The respondents skipped the Explainer videos **more than 2 times less often** than expected.
4. **Test score.** After watching the videos, the respondents performed **more than two times better** than on the initial test.
5. **SUS.** On the SUS, respondents mostly **agreed or strongly agreed** with positive statements about the videos.

The videos were characterised as **informative, useful, understandable, concise, engaging, and visually appealing**.

11.5.1 Goal achievement

Based on those KPIs and general feedback from the respondents, it is safe to say that all the **goals of the project were successfully achieved**.

12. Distribution

First, the videos should be uploaded to a streaming service like YouTube. In the case of animated videos, icon creators need to be credited in the video description. A list of those attributions can be found in the Appendix.

The recommended channel for further distribution is the VŠE website. Separate web pages for different tutorials should be created according to the qualitative research, during which the participants expressed confusion because of too much information on a single web page. New videos should be opened in new tabs so that the users can easily go back to the previous videos.

All the Tutorial videos feature a disclaimer at the beginning of the video which was found necessary in the Customer Journeys and during the Usability testings. Nonetheless, if those disclaimers are found too long or inessential, they can be skipped by embedding the videos using a time code, so that the videos would play from a certain point.

Every video description on the website should feature **all links mentioned in the video**. Also, **links to the other videos** should be included as presented in the chart below. It is also advisable to include QR codes leading to the page itself in case of tutorials for operating systems Windows, macOS and Linux since users would be likely to open those videos on their mobile devices. Links needed in the videos can be found in Appendix.

It is also advisable to include a downloadable text file with a full video transcript to accommodate users who would prefer a text version.

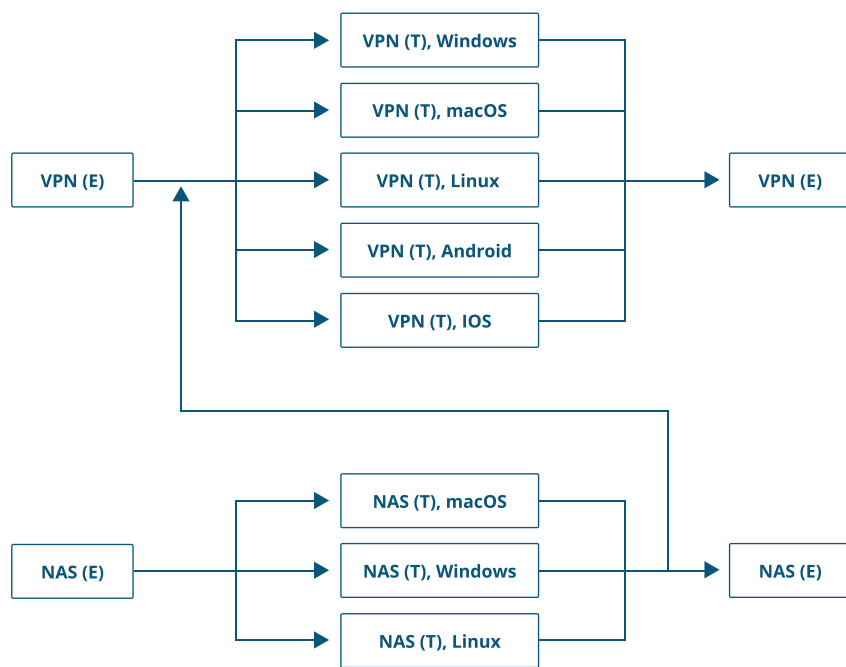


Table 12.1: Distribution

13. Next steps

13.1 Promotion

It is advisable to introduce the videos to the target audience **before** they need to access the school VPN or NAS so that they know where to look for instructions on how to access the services needed when they actually need them. That way, without an immediate need to connect to the service, students could be more likely to watch explainer videos. That may be especially successful if information about the videos is emailed to them at the beginning of their journey at VŠE.

13.2 Burned-in English captions

To avoid additional confusion, it is possible to create videos with burned-in English captions placed on a separate web page in English. That way, even though the voiceover was in Czech, it would be immediately clear that the video is intended for English speakers.

13.3 Gamification: multiple-choice tests

In the Final Product Validation phase, participants were asked to assess their knowledge of NAS and VPN before and after watching the Explainer videos. Those tests were in form of open questions and therefore had to be moderated. The respondents expressed that it was a good idea to incorporate multiple-choice tests next to the videos because that made them reflect on how well they understood the video and rewatch the video to give the right answers. If their answers were not correct, they could review the material by seeing the right ones. The other reason expressed by the participants was that once they knew that there was a test after the video, it looked like a challenge to hear all the necessary information to give the right answers; therefore it felt more like a game.

13.4 Alternative authentication method

There are two possible authentication methods to access the VŠE VPN:

1. Logging in by the Eduroam login and password.
2. Authentication by a digital certificate.

In the tutorials explaining how the school VPN can be accessed, authentication with the Eduroam login and password was described, leaving out authentication using a digital certi-

ificate. That instruction already exists on a VŠE web page but seeing that in the research it was found that the instructions in the form of a video worked better, it could be also remade as a video instruction.

If that video manual is created, it is important not to overcomplicate the Customer Journey, possible problems being:

- authentication with a digital certificate may take several days and therefore is not suitable for solving an immediate need,
- users may be frustrated by too many options on how to authenticate if they cannot tell the difference between them,
- users may think that two authentication methods are two necessary steps if they are placed subsequently on each page.

A possible solution to those problems might be the following:

1. Keep the Customer Journey map as it is and let the users solve an immediate need.
2. Avoid any choice of authentication method.
3. Once they have solved their problem by accessing the school VPN, provide them with the possibility of authenticating differently next time by including a link to the video explaining how to do that. In the Customer Journey map in this project, that would correspond to the step Care.
4. It is important to highlight the advantages and disadvantages of authenticating using different methods so that the user can make an educated decision.

14. Timing and Budget

The project was more time-consuming than expected. The problem Discovery and Problem Definition phases took more time than originally planned. Then, for the Concept Validation phase, respondents had to be gathered. Collaborating with people took more time than expected and therefore prolonged the project. On the other hand, participants taking part in the Final Product Validation Phase were recruited in advance and therefore everything went according to plan. Video editing and animation (nearly 25 minutes in total) took more time than originally expected due to the results of Concept Validation, where it was found that recapturing the screen could benefit the project. In total, all steps took approximately 135 hours.

It was possible to create this project almost without investing anything, apart from human resources. Studio for the sound recording was provided by VŠE as well as recording hardware and software. While capturing screencast either open-source free or default (and thus free) software was used. The participants took part in the project without demanding remuneration as well. The only fee actually paid was the student subscription to Adobe Creative Cloud for 20 euros monthly.

Had this not been a student project, it could have cost nearly 40 000 CZK, not counting production and over 100 000 CZK in total.

Production timing	Time, hours	Possible Budget	Price, CZK
Video editing	35	Research participants	10 000
Animating	30	Voiceover actor	7 500
Text creation	30	Sound recording studio	4 500
User testing	25	Software	8 700
Researches	20	Testing participants	15 000
Audio recording	12	Music	5 000
Validations	4	Subtotal	50 700
Audio editing	10	Production	60 000
Total	161	Total	110 700

Table 14.1: Timing and Budget

15. Conclusion

Although designing a product using HCD takes more time than its development based on designers' ideas of users' needs and frustrations, in this thesis it proved to be worthy of the time spent on research, testing, and iteration. Moreover, a product developed without prior research and user testing before launch might later need a complete redesign. That is usually less cost-effective than creating a product intended for users in the first place. That is especially important if the product is supposed to generate income since a product that does not satisfy users' needs will not make the same profits as one that does.

All the steps in the Triple Diamond design process model were found to be exceptionally useful and important. In the first phase, it was deduced that although there already were some tutorials about accessing VPN and NAS, there was indeed demand for a different type of tutorial on that topic. Thanks to the qualitative research conducted in the second phase of this project, it was clear that users did not only need to know how to access VPN and NAS but also an explanation of how they worked. In the third phase, video types and KPIs were established, as well as challenges and edge cases that helped create tutorials users would appreciate. The Concept Validation phase ensured the project was advancing in the right direction and helped detect and fix minor problems in the videos, which could be harder to fix in the final product. In Design and Development phase, the videos were created and edited with regard to edge cases and users with disabilities. Final Product Validation was necessary to determine whether the created videos satisfied users' needs and how they made them feel overall.

The outcome of this thesis, 10 video tutorials, did prove to satisfy users' needs and solve potential users' problems in the Final Product Validation phase. Even if the users made mistakes while following the process described in the video, they were able to solve the problem only by using the video itself. That was achieved by highlighting important and often confusing information in the form of an animation or text overlay on the screen.

The project benefits all KME students at VŠE and all students who need a VŠE VPN.

Acronyms

AIDA Awareness–Interest–Desire–Action. 5, 36

FPS frames per second. 20, 44, 46, 51

HCD Human-centered design. 1, 3, 67

HD High Definition. 44, 51

HMW How might we?. 5, 32

KME Multimedia department. 1, 25, 28, 33, 37, 67

KPIs Key Performance Indicators. 1, 10, 35, 53–55, 59, 67

NAS Network Attached Storage. 1, 18, 25–35, 37, 38, 41, 48, 49, 54, 55, 57–59, 63, 67

STDC See–Think–Do–Care. 5, 36

SUS System Usability Scale. 10, 54, 55, 57, 59

UHD Ultra High Definition. 44, 46, 51

VPN Virtual Private Network. 1, 17–19, 25–38, 41, 43, 46, 49, 54–59, 63, 64, 67

VŠE The Prague University of Economics and Business. 1, 18, 23, 25, 27–30, 33, 38, 40, 45, 49, 55, 61, 63–65, 67

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A. Video Scripts

A.1 VPN videos

A.1.1 Czech version

VPN (E), Co je VPN?

VPN, neboli Virtual Private Network, v češtině Virtuální Privátní Síť, zaručuje bezpečný přenos dat přes nezabezpečený internet.

VPN vytváří zabezpečené spojení mezi dvěma místy v síti. To znamená, že zařízení, ze kterého se připojujete, a zařízení, ke kterému se připojujete, dokážou vytvořit mezi sebou zabezpečený tunel a ten Vám umožní posílat a přijímat informace přes internet tak, že nikdo jiný Vaše data nepřečte.

Možná, že VPN již používáte, abyste například obešli geoblokaci, když se chcete podívat na pořad, který není v České republice dostupný, nebo chcete chránit své soukromí a používáte VPN pro snížení množství informací, které se o Vás mohou ostatní dozvědět.

Proč ale VPN použít pro připojení do školy?

Určitě znáte areál na Žižkově. Tady se nachází servery, na kterých je uloženo značné množství studentských dat. Ta nejsou vně organizace dostupná a to především z bezpečnostních důvodů.

Nebylo by přece moudré, aby měl kdokoliv odkudkoliv vzdálený přístup k jakémukoliv počítači ve školní síti.

To je důvod, proč musíme použít VPN, která navíc před odesláním všechna data zašifruje tak, aby je mohl dešifrovat jenom recipient - teda příjemce.

Takže i kdyby někdo Vaši komunikaci se školou sledoval, bez klíče data dešifrovat nedokáže, a jsou pro něj tudíž bezcenná.

Toto se hodí studentům nebo zaměstnancům, kteří se potřebují dostat k datům uloženým ve škole nebo potřebují použít virtuální prostředí Horizon či specifické aplikace a weby používané pro výuku.

VPN pro Vzdálený Přístup, anglicky Remote Access, připojí Vaše zařízení a to bez ohledu na to, odkud jste na internet připojení.

Pro připojení k VPN potřebujete klientskou aplikaci, jako je například Cisco Anyconnect, kterou použijete, abyste se připojili do sítě VŠE.

Tato aplikace Vás připojí do školní sítě, pokud se prokážete platným uživatelským jménem a heslem, které používáte pro připojení do Eduroamu.

Rovněž se můžete připojit pomocí certifikátu.

Připojení pomocí VPN vlastně předstírá, že jste fyzicky připojení přímo ve školní síti.

VPN tak není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Připojením na školní VPN využijete Rozděleného Tunelování, anglicky Split Tunnelling.

To znamená, že VPN se použije pouze na spojení do sítě VŠE, ale nijak neovlivní Vaše připojení do zbytku internetu. Proto školní VPN nelze využít pro obejítí geoblokace či zachování anonymity.

Takže, kdyby se student či zaměstnanec chtěl připojit z jakéhokoliv zařízení do školní sítě odkudkoliv mimo síť VŠE, použije aplikaci Cisco AnyConnect, která mu po úspěšném přihlášení vytvoří tunel mezi jeho zařízením a školou.

Tím tak získá přístup k požadovaným datům a službám, jako kdyby se nacházel ve škole.

To, jak nastavit VPN pro Vaše zařízení, se dozvíte v následujícím videu.

VPN (T), Windows

VPN není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Pokud používáte operační systém Microsoft Windows, na školní VPN se připojíte jednoduše.

Prvním krokem je instalace klienta, což je aplikace Cisco AnyConnect, kterou si stáhnete z odkazu pod videem nebo když do adresního řádku prohlížeče zadáte „vpn.vse.cz“.

Na nově otevřené stránce se musíte přihlásit.

Potřebujete stejné údaje, jako pro přihlášení na školní WIFI, teda školní emailovou adresu či xname a heslo do Eduroamu, nikoli do InSISu.

Až se přihlásíte, stáhněte si aplikaci. Nainstalujte si ji jako obvykle.

Druhým krokem aplikaci spusťte.

Do pole vedle Connect zadejte adresu školního VPN serveru „vpn.vse.cz“ a klikněte na Connect.

V novém okně se potřebujete přihlásit ještě jednou.

Potřebujete stejné údaje jako v předchozím kroku, teda školní emailovou adresu či xname a heslo do Eduroamu.

Pro připojení klikněte na ok.

To je vše! Nyní jste připojeni na školní VPN.

Pokud se znovu potřebujete přihlásit, jednoduše si zopakujte druhý krok.

VPN (T), MacOS

VPN není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Pokud používáte operační systém MacOS, na školní VPN se připojíte jednoduše.

Prvním krokem je instalace klienta, což je aplikace Cisco AnyConnect, kterou si stáhnete z odkazu pod videem nebo když do adresního řádku prohlížeče zadáte „vpn.vse.cz“. Na nově otevřené stránce se musíte přihlásit.

Potřebujete stejné údaje, jako pro přihlášení na školní WIFI, teda školní emailovou adresu či xname a heslo do Eduroamu, nikoli do InSISu.

Až se přihlásíte, stáhněte si aplikaci. Nainstalujte si ji jako obvykle.

Po instalaci se objeví oznámení o blokování rozšíření.

Klikněte dole na Otevřete Předvolby, v novém okně vlevo dole klikněte na zámek a zadejte heslo uživatele počítače.

Poté povolte načtení systémového softwaru z aplikace Cisco AnyConnect kliknutím na tlačítko Povolit.

Znovu klikněte na zámek pro zamezení dalších změn a zavřete všechna zbytečná okna.

Druhým krokem aplikaci spusťte.

Do pole vedle Connect zadejte adresu školního VPN serveru „vpn.vse.cz“ a klikněte na Connect.

V novém okně se potřebujete přihlásit ještě jednou.

Potřebujete stejné údaje jako v předchozím kroku, teda školní emailovou adresu či xname a heslo do Eduroamu.

Pro připojení klikněte na ok.

To je vše! Nyní jste připojeni na školní VPN.

Pokud se znovu potřebujete přihlásit, jednoduše si zopakujte druhý krok.

VPN (T), Linux

VPN není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Prvním krokem pro připojení na školní VPN na Linuxu v prostředí Gnome je instalace balíčku VPN klienta pomocí příkazu, který je k nalezení pod videem.

Otevřete terminál kliknutím na Činnosti a vyhledáním terminálu. Také můžete použít klávesovou zkratku Control + Alt + T.

Do Terminálu vložte příslušný příkaz pro Váš systém, a to pomocí klávesové zkratky Control + Shift + V a zadejte heslo uživatele počítače.

Dotaz, zda chcete pokračovat, stačí potvrdit klávesou Enter.

Po dokončení instalace zavřete Terminál.

Po instalaci programu otevřete Nastavení v menu vpravo nahoře.

Pak zvolte Síť a vpravo od názvu sekce VPN klikněte na tlačítko se symbolem plus.

Zvolte víceprotokolový klient VPN a otevře se okno s nastaveními sítě.

Připojení můžete pojmenovat například VŠE.

Ujistěte se, že zvolený protokol VPN je Cisco Anyconnect.

Do pole Brána vložte adresu „vpn.vse.cz“ a klikněte na tlačítko Přidat.

Kliknutím na přepínač zapněte v nastavení nově vytvořenou VPN a přihlaste se.

Potřebujete stejné údaje, jako pro přihlášení na školní WIFI, teda školní emailovou adresu či xname a heslo do Eduroamu, nikoli do InSISu.

Pokud chcete, zaškrtněte volbu pro uložení hesla a klikněte na Login.

To je vše!

Vpravo nahoře je ikonka VPN, která po odpojení zmizí. Pokud se znovu potřebujete připojit na školní VPN, jednoduše připojení aktivujte v menu v pravém horním rohu.

VPN (T), iOS

VPN není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Pokud používáte iPhone, je nutné si nainstalovat aplikaci Cisco Secure Client z AppStore, kterou si můžete stáhnout z odkazu pod videem nebo když vyhledáte Cisco Secure Client.

Po otevření aplikace zvolte, jestli chcete dostávat upozornění a klikněte na Připojení a Přidat připojení k síti VPN.

V novém okně zadejte Popis sítě, například VŠE. Jedná se pouze o název, který nemá na připojení vliv.

V druhém políčku zadejte adresu serveru „vpn.vse.cz“. Provedené nastavení uložte.

V novém dialogu povolte aplikaci monitorování síťové aktivity a zadejte heslo do Vašeho telefonu.

Poté se vraťte na úvodní obrazovku a zapněte Vámi nově nastavenou VPN.

Nyní se musíte přihlásit.

Potřebujete stejné údaje, jako pro přihlášení na školní WIFI, teda školní emailovou adresu či xname a heslo do Eduroamu, nikoli do InSISu. Klikněte na Připojit.

Nyní se Vám zobrazí úvodní obrazovka Se stavem připojení.

Pokud se budete chtít odhlásit, stačí vypnout přepínač vedle názvu AnyConnect VPN.

Pokud se znovu potřebujete připojit na školní VPN, stačí VPN přepínačem zapnout a znovu se přihlásit.

VPN (T), Android

VPN není třeba, pokud se připojujete ze sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci.

Pokud používáte Android, je nutné si nainstalovat aplikaci Cisco Secure Client z Google Play, kterou si můžete stáhnout z odkazu pod videem nebo když vyhledáte Cisco Secure Client.

Po otevření aplikace přijměte licenční smlouvu a povolte aplikaci Volat a spravovat telefonní hovory. Pro nastavení VPN klikněte na Připojení a tlačítko se symbolem plus vpravo dole.

V novém okně zadejte Popis sítě, například VŠE. Jedná se pouze o název, který nemá na připojení vliv.

V druhém políčku zadejte adresu serveru „vpn.vse.cz“. Provedené nastavení uložte.

Provedené nastavení uložte kliknutím na tlačítko Hotovo.

Poté se vraťte na úvodní obrazovku a zapněte Vámi nově nastavenou VPN.

Nyní se musíte přihlásit.

Potřebujete stejné údaje, jako pro přihlášení na školní WIFI, teda školní emailovou adresu či xname a heslo do Eduroamu, nikoli do InSISu.

Klikněte na Připojit.

V novém dialogu povolte aplikaci sledovat síťový provoz.

Nyní se Vám zobrazí úvodní obrazovka se stavem připojení.

Pokud se budete chtít odhlásit, stačí vypnout přepínač vedle názvu AnyConnect VPN.

Pokud se znovu potřebujete připojit na školní VPN, stačí VPN přepínačem zapnout a znovu se přihlásit.

A.1.2 English version

VPN (E), What is VPN?

VPN, or Virtual Private Network, Virtuální Privátní Sít' in Czech, guarantees secure data transmission when connected to unsecured network.

A VPN creates a secure connection between two locations in a network. This means that the device you're connecting from and the device you're connecting to can create a secure tunnel, allowing you to send and receive information over the Internet without anyone else reading your data.

Perhaps you have already used a VPN to bypass geoblocking, when you wanted to watch a show that is not available in the Czech Republic, or you want to protect your privacy and use a VPN to reduce the amount of information that others can learn about you.

But why use a VPN to connect to school?

You certainly know the campus in Žižkov. This is where the servers are located, on which a significant amount of student data is stored. It is not available outside the organization, primarily for security reasons.

It would not be wise to let everyone have a possibility to remotely access any computer in the school network.

This is the reason why we have to use a VPN, which also encrypts all data before sending it, so that only the recipient - that is, the receiver - can decrypt it.

Even if someone monitored your communication with the school, without the key, the data cannot be decrypted, and it isn't therefore valuable.

VPN is used by students and staff who need to access data stored in the school or need to use the Horizon virtual environment or specific applications and websites used in school.

Remote Access VPN connects devices from any network.

To connect to the VPN, you need a client application, such as Cisco AnyConnect, which is used to connect to the VŠE network.

This application will connect you to the school network if you login with a valid username and password that you use to connect to Eduroam.

You can also connect using a certificate.

While connected, VPN actually pretends that you are physically connected to the school network.

A VPN is therefore not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichuv Hradec.

School VPN provides Split Tunneling connection.

This means that the VPN will only be used to connect to the VŠE network, but will not affect your connection to the rest of the Internet in any way. Therefore, the school VPN cannot be used to bypass geoblocking or to protect data.

So, if a student or employee connected to any network wants to connect from any device to the school network, they will use the Cisco AnyConnect application, which, once user has logged in, will create a tunnel between their device and the school.

This way, they will get access to the required data and services, as if they were at school physically.

You will learn how to set up a VPN for your device in the following video.

VPN (T), Windows

A VPN is not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichov Hradec.

If you use the Microsoft Windows operating system, connecting to the school VPN is rather easy. The first step is installing the client, which is the Cisco AnyConnect application. You can download it from the link below the video or by entering “vpn.vse.cz” in the address bar in your browser.

You must log in on the newly opened page.

Here you will need the same credentials as for the school WIFI: the school email address or xname and password to Eduroam, not to InSIS. Once you're logged in, please download the app. Install it as you do usually.

The second step is launching the application.

In the field next to Connect, enter the school VPN server address “vpn.vse.cz” and click Connect.

You need to log in again in the new window.

You will need the same credentials as in the previous step: school email address or xname and password to Eduroam. Click ok to connect.

That's all! You are now connected to the school VPN. If you need to sign in again, simply repeat the second step.

VPN (T), MacOS

A VPN is not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichov Hradec.

If you use the MacOS operating system, connecting to the school VPN is rather easy.

The first step is installing the client, which is the Cisco AnyConnect application. You can download it from the link below the video or by entering "vpn.vse.cz" in the address bar in your browser.

You must log in on the newly opened page.

Here you will need the same credentials as for the school WIFI: the school email address or xname and password to Eduroam, not to InSIS.

Once you're logged in, download the app. Install it as you do usually.

Now a notification about blocking the extension will appear.

Click on Open Preferences at the bottom. Then in the new window, click on the lock.

Enter user's password.

Then, allow system software monitoring by clicking the Allow button.

Click the lock to prevent further changes and close unnecessary windows.

The second step is launching the application.

In the field next to Connect, enter the school VPN server address "vpn.vse.cz" and click Connect.

You need to log in again in a new window.

You will need the same credentials as in the previous step: school email address or xname and password to Eduroam. Click ok to connect.

That's all! You are now connected to the school VPN.

If you need to sign in again, simply repeat the second step.

VPN (T), Linux

A VPN is not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichov Hradec.

The first step to connect to a school VPN on Linux in Gnome is installing the VPN client

package using the command below the video.

Open Terminal by clicking on Activities and searching for Terminal. You can also use the keyboard shortcut Control + Alt + T.

In Terminal, enter the appropriate command for your system using the keyboard shortcut Control + Shift + V and enter the user's password.

Confirm the question whether you want to continue by pressing Enter.

Once the installation is completed, close Terminal.

After installation open Settings in the menu at the top right corner.

Choose Network and click the plus button to the right of the VPN section.

Choose a multi-protocol VPN client.

You can name the connection VŠE.

Make sure the selected VPN protocol is Cisco AnyConnect.

Enter the address "vpn.vse.cz" in the Gateway field and click the Add button.

Click the switch to turn on the newly created VPN in settings and log in.

You will need the same credentials as for school WIFI: the school email address or xname and password to Eduroam, not to InSIS.

If you want, select the checkbox to save the password and click on Login.

That's all!

VPN icon at the top right corner will disappear after disconnection. If you need to connect to the school VPN again, simply activate the connection in the menu in the upper right corner.

VPN (T), iOS

A VPN is not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichuv Hradec.

If you are using an iPhone, you must install the Cisco Secure Client app from the AppStore. You can find the link below the video or by searching for Cisco Secure Client.

After opening the app, choose if you want to receive notifications and click Connect and Add VPN Connection.

In the new window, choose a Description of the network. It does not affect the connection.

In the second field, enter the server address "vpn.vse.cz". Save the settings.

In the new dialog, allow network activity monitoring and enter the password in your iPhone.

Go back to the home screen and turn on your newly set up VPN.

Now you will need to log in.

You will need the same credentials as for the school WIFI: school email address or xname and password to Eduroam, not to InSIS. Click Connect.

You will now see the connection status on the home screen. If you want to log out, just turn off the switch next to the AnyConnect VPN.

If you need to connect to the school VPN again, just turn the VPN switch on and log in again.

VPN (T), Android

A VPN is not needed if you are connecting from the VŠE network, their dormitories or from the Eduroam wireless network in the VŠE buildings in Prague and Jindřichuv Hradec.

If you are using Android, you must install the Cisco Secure Client app from Google Play. You can find the link below the video or by searching for Cisco Secure Client.

After opening the app, accept the license agreement and allowing managing phone calls.

To set up a VPN, click Connections and the plus button at the bottom right.

In the new window, choose a Description of the network. It does not affect the connection.

In the second field, enter the server address "vpn.vse.cz". Save the settings by clicking the Done button.

Go back to the home screen and turn on your newly set up VPN. Now you will need to log in.

You will need the same credentials as for the school WIFI: school email address or xname and password to Eduroam, not to InSIS. Click Connect.

In the new dialog, allow the application monitoring network traffic.

You will now see the connection status on the home screen.

If you want to log out, just turn off the switch next to the AnyConnect VPN.

If you need to connect to the school VPN again, just turn the VPN switch on and log in again.

A.2 NAS videos

A.2.1 NAS (E), Co je NAS?

NAS je v podstatě školní síťové úložiště.

NAS, neboli anglicky Network Attached Storage je datové úložiště, které dovoluje uživatelům z více zařízení nahrávat, uchovávat, a stahovat data.

NAS se skládá buď z jednoho, ale spíše z více pevných disků, umístěných v jednom zařízení které tvoří diskové pole s vysokou datovou kapacitou. Na rozdíl od externích disků, se NAS připojuje přímo do počítačové sítě.

Všechna zařízení v dané síti se tak dostanou k datům, která jsou na něm uložena.

To je důvod, proč pro přístup ke školnímu NASu potřebujete připojení ke školní síti: buď fyzické, nebo pomocí VPN.

Díky školnímu NASu jsou všechny studentské výstupy bezpečně uchovávány na jednom místě tak, že oprávněný uživatel může prohlížet, stahovat a nahrávat data kdykoliv a odkudkoliv.

Z bezpečnostních důvodů mají jednotliví uživatelé různá oprávnění.

Například, aby student omylem nesmazal svůj či cizí soubor, nemá povolení soubory mazat. To znamená, že když svůj soubor nahrajete, smazat ho dokáže jenom učitel, i když je to Váš soubor.

Prohlížet a nahrávat soubory může student jenom v individuálně zpřístupněných složkách, proto se do NASu musíte přihlásit pomocí přihlašovacích údajů, které používáte do InSISu: teda pomocí uživatelského jména - neboli xname (například, mark04) a hesla.

Na NASu se nachází dvě sdílené složky: Odevzdávárna a Výuka. Do odevzdávárny budete požadované výstupy nahrávat. To ale můžete jenom do složek předmětů, které studujete v aktuálním semestru.

Ve Výuce se nachází všechny studijní materiály jako prezentace, práce z minulých let a další. Prohlížet Vám dostupné soubory můžete bez omezení. Soubory ale nemohou studenti upravovat či mazat.

To, jak se připojit na NAS z Vašeho zařízení, najdete v následujícím videu.

NAS (T), Windows

Když nejste připojení do sítě VŠE a jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci, potřebujete se do ní připojit přes VPN.

Odkaz na návod je k nalezení pod videem.

Abyste se připojili na NAS, potřebujete nejdříve otevřít Průzkumníka.

Takže, klikněte na Start a poté na ikonku Průzkumníka, nebo můžete použít klávesovou zkratku Windows + E.

Klikněte pravým tlačítkem myši na Tento počítač, zvolte Více možností a klikněte na Přidat síťovou jednotku.

V novém okně do políčka Složka zadejte požadovanou adresu, která je k nalezení pod videem.

Zkontrolujte, že jsou pole „Znovu připojit při přihlášení“ a „Připojit pomocí jiných přihlašovacích údajů“ zaškrtnutá. Pak klikněte na Dokončit.

Pro přihlášení potřebujete použít přihlašovací údaje do InSISu.

Do pole uživatelské jméno zadejte velká písmena AD, zpětné lomítko a Váš xname, což jsou 4 písmena a dvě čísla jako ve školním emailu.

Použijte heslo do InSISu.

Písmena AD jsou nutná, aby server úspěšně vyhledal Vaše přihlašovací údaje.

Pokud chcete, zvolte Zapamatovat přihlašovací údaje a klikněte na ok.

To je vše! Nyní jste připojení na NAS a můžete prohlížet a nahrávat soubory.

NAS (T), MacOS

Když nejste připojení do sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci, potřebujete se do ní připojit přes VPN, odkaz na návod je k nalezení pod videem.

Abyste se připojili na NAS, klikněte na panel Otevřít v menu vlevo nahoře a zvolte Připojit k serveru. Také můžete použít zkratku Command + K.

V novém okně zadejte adresu serveru, kterou naleznete pod videem. Klikněte na Připojit.

Zkontrolujte, že se přihlašujete jako Registrovaný uživatel a použijte přihlašovací údaje do InSISu.

Do pole jméno zadejte velká písmena AD, zpětné lomítko a Váš xname, což jsou 4 písmena a dvě čísla jako ve školním emailu.

Použijte heslo do InSISu.

Písmena AD jsou nutná, aby server úspěšně vyhledal Vaše přihlašovací údaje.

Pokud chcete, zaškrtněte volbu pro uložení hesla do Svazku klíčů.

Klikněte na Připojit a zvolte složku, kterou chcete připojit.

Můžete zvolit i více složek stisknutím klávesy Command.

Klikněte na ok a to je vše! Nyní jste připojení na NAS a můžete prohlížet a nahrávat soubory.

NAS (T), Linux

Když nejste připojení do sítě VŠE, jejich kolejí anebo z bezdrátové sítě Eduroam v budovách VŠE v Praze a Jindřichově Hradci, potřebujete se do ní připojit přes VPN, odkaz na návod je k nalezení pod videem.

Abyste se připojili na NAS, potřebujete nejdříve otevřít Soubory.

Klikněte vlevo nahoře na Činnosti nebo použijte klávesu Super.

A poté dole uprostřed klikněte na ikonku aplikace Soubory.

Vlevo pod složkami klikněte na Další umístění a dole se Vám zobrazí Připojit k serveru.

Do políčka vpravo zadejte požadovanou adresu, která je k nalezení pod videem.

Klikněte na Připojit.

V novém okně zkontrolujte, že se přihlašujete jako registrovaný uživatel.

Použijte přihlašovací údaje do InSISu.

Do pole uživatelské jméno zadejte Váš xname, což jsou čtyři písmena a dvě čísla jako ve školním emailu,

Do políčka Doména napište velká písmena AD.

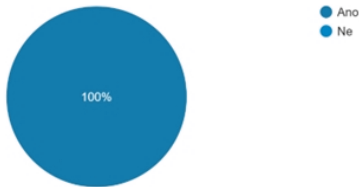
Použijte heslo do InSISu.

Pokud chcete, zaškrtněte volbu pro uložení hesla a klikněte na Připojit.

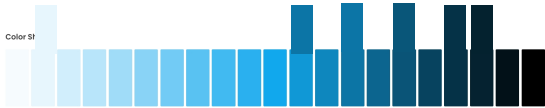
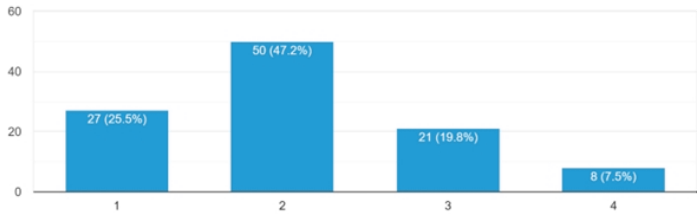
To je vše! Nyní jste připojení na NAS a můžete prohlížet a nahrávat soubory.

B. Results of the quantitative research

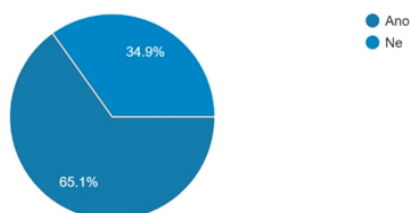
Jsem student VŠE
106 responses



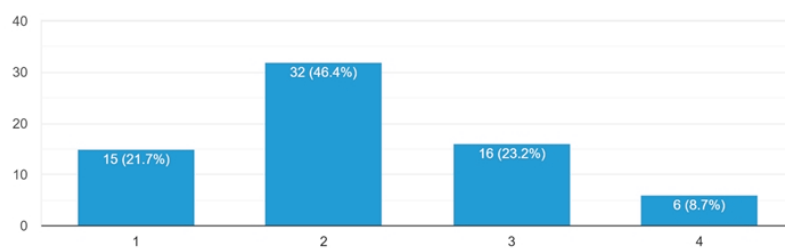
Jak technicky zdáný(á) jsi
106 responses



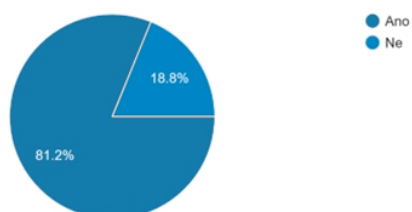
Pro své studium na VŠE jsem se potřeboval(a) připojit na školní VPN
106 responses



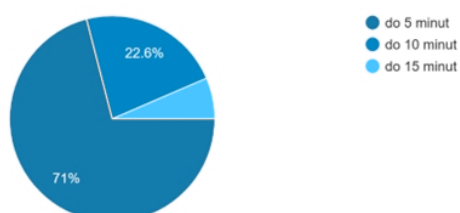
Jak ti šlo připojení na školní VPN?
69 responses



Pro připojení na školní VPN bych uvítal(a) video návod
69 responses

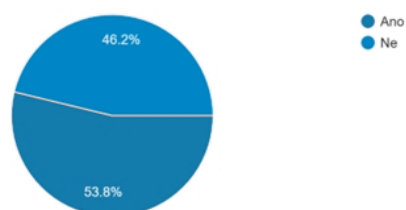


Jak dlouhý video návod by sis pustil(a)?
62 responses



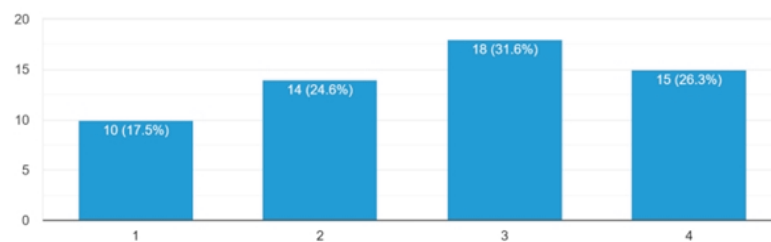
Pro své studium na VŠE jsem se potřeboval(a) připojit na datové úložiště NAS (síťový disk vyuka nebo odevzdavarna)

106 responses



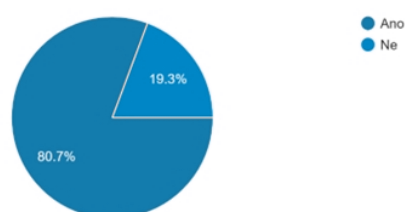
Jak ti šlo připojení na školní NAS?

57 responses



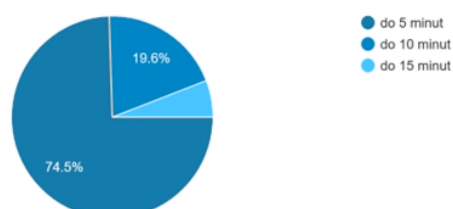
Pro připojení na školní NAS bych uvítal(a) video návod

57 responses



Jak dlouhý video návod by sis pustil(a)?

51 responses



C. Concept Validation Script

1. Before watching VPN (E) and VPN (T) videos

- How do you access your technology skills?
- Do you use VPN? What for?
- How well do you know how VPN works?
- If I told you to access a certain server, what would you do?

2. After watching VPN (E) video

- Has your understanding of VPN changed? How?
- In what circumstances do you think you would need this video?
- Was there something that was hard to understand in the video?
- How do you feel about the visuals?
- How did you feel about the voiceover?

3. After watching VPN (T) videos

- In what circumstances would you use that video?
- If you had to access school VPN, what would you do?
- Was there something that was hard to understand in the video?
- How do you feel about the visuals?
- Is there anything you would like to add?

4. Before watching NAS (E) video

- Have you heard about NAS?
- Do you know how it works?
- If you had to connect to it, what would you do?

5. After watching NAS (E) video

- Has your understanding of NAS changed? How?
- In what circumstances do you think you would need this video?
- Was there something that was hard to understand in the video?
- How do you feel about the visuals?
- How did you feel about the voiceover?

6. After watching NAS (T)

- In what circumstances would you use that video?
- If you had to access NAS, what would you do?
- Was there something that was hard to understand in the video?
- How do you feel about the visuals?

7. Additional information

- How do you feel about the videos?
- Do you think you would make use them?
- What would you change in the videos?
- Is there anything you would like to add?

D. Final Product Validation Script

1. Before watching videos — Introduction

Hello, thank you so much for participating in this testing. First, I will ask you some general questions about your experience.

- How do you access your technology skills?
- Do you use a VPN? What for?

2. Before watching VPN (T) videos — Task explanation

Your goal is to connect to a VŠE VPN. There are several videos that you can access, the first one of them gives an overall idea of how a VPN works, while the others are practical manuals made for specific platforms. Also, you can access a written manual.

Right now, you have a device with a Windows/MacOS/Linux/Android/iOS (choose applicable) operating system installed. To access VPN, you will receive two sets of credentials. One of them you can use to connect to the Eduroam network, the other one is used to access InSIS, Intranet, and other VŠE applications. I would ask you to comment while you are performing actions and generally act as if you would normally. Do you have any questions?

3. After watching VPN (T) videos — Feedback

- How do you feel about watched videos? Did you encounter any problems while accessing VPN?
- Would you use them to connect to a VPN next time? Why? Why not?
- On scale from 1 to 10 where 1 is “never” and 10 “certainly”, how likely are you to recommend those videos to a classmate if they ask you for advice on a VPN connection to the school network?

4. Before watching VPN (E) video

Before watching next video, I would like to ask you some question to have an idea of your understanding how VPN works.

1. In your own words, what is a VPN?
2. Do you know why VPN is needed to access school services?
3. Do you have an idea how your personal information is secured while you are connected to a VPN?
4. Do you know what a Remote Access VPN means?
5. Do you know what Split Tunneling connection means?

Now you may watch a video which gives an overall idea of how VPN works and how it is used at VŠE. After having watched it you will be asked some questions on that topic. You do not have to watch it if you feel you will do good on the test.

5. Test — After watching VPN (E) video

Now I will ask you the same questions concerning VPN that I posed before. If you do not remember something, that is alright, this is not a test to access your knowledge.

1. In your own words, what is a VPN?
2. Do you know why VPN is needed to access school services?
3. Do you have an idea how your personal information is secured while you are connected to a VPN?
4. Do you know what a Remote Access VPN means?
5. Do you know what Split Tunneling connection means?

6. Usability Scale — After watching VPN (E) video

Thank you for your answers; now, could you please express your experience using those statements? For each one, you can choose if you strongly agree, strongly disagree, agree, disagree, or not sure.

1. I think that I would like to watch this video when I want to refresh my knowledge of how the school VPN works.
2. I did not find the video too complex.
3. I thought the video was easy to understand.
4. I do not think that I would need the support of a technical person to be able to understand the notions explained in the video.
5. I found the icons, music, and voice to be well integrated.
6. I did not think there was too much inconsistency in this video.
7. I would imagine that most people would understand this video well.
8. I did not find the video unstructured.
9. I found the voiceover nice to listen to.
10. I felt very confident about my knowledge of notions explained in the video after having watched it

15 minutes break

7. Before watching NAS (T) videos — Task explanation

Now we will proceed to the other half of our meeting. Here is a list of all links and commands mentioned in the videos. Your current goal is to connect to NAS. Just as previously, there are several videos that you can access, the first one of them gives an overall idea of how NAS works, the others are practical manuals made for specific platforms. Also, you can access a written manual. Right now you have a device with Windows/MacOS/Linux/ (choose applicable) operating system installed.

To access NAS, you may use credentials that you have already received. I would ask you to comment while you are performing actions and generally act as if you would normally. Do you have any questions?

8. After watching NAS (T) videos — Feedback

- How do you feel about those videos? Did you encounter any problems while accessing NAS?
- Would you use them to connect to NAS next time? Why? Why not?
- On scale from 1 to 10 where 1 is “never” and 10 “certainly”, how likely are you to recommend those videos to a classmate if they ask you for advice on connecting to NAS?

9. Before watching NAS (E) video

Before watching next video, I would like to ask you some question to have an idea of your understanding how NAS works.

1. In your own words, what is NAS?
2. Do you know why to access NAS you need to access VPN or be directly connected to school network?
3. Do you know why you have to use InSIS credentials in order to access NAS?
4. Do you know which credentials you need to access NAS and why?
5. Do you know which permissions a student has for folders in NAS?

Now you may watch a video which gives an overall idea of how VPN works and how it is used at VŠE. After having watched it you will be asked some questions on that topic. You do not have to watch it if you feel you will do good on the test.

10. Test — After watching NAS (E) video

Now I will ask you the same questions concerning NAS that I posed before. If you do not remember something, that is alright, this is not a test to access your knowledge.

1. In your own words, what is NAS?
2. Do you know why to access NAS you need to access VPN or be directly connected to school network?
3. Do you know why you have to use InSIS credentials in order to access NAS?
4. Do you know which credentials you need to access NAS and why?
5. Which permissions does a student have for folders Odevzdavarna and Vyuka?

11. Usability Scale — After watching NAS (E)

Thank you for your answers, now please could you express your experience using those statements? To each one you can choose if you strongly agree, strongly disagree, agree, disagree or prefer not to say.

1. I think that I would like to watch this video when I want to refresh my knowledge of how the school NAS works.
2. I did not find the video too complex.
3. I thought the video was easy to understand.
4. I do not think that I would need the support of a technical person to be able to understand the notions explained in the video.
5. I found the icons, music, and voice to be well integrated.
6. I did not think there was too much inconsistency in this video.
7. I would imagine that most people would understand this video well.
8. I did not find the video unstructured.
9. I found the voiceover nice to listen to.
10. I felt very confident about my knowledge of the notions explained in the video after having watched it.

12. Additional information

- Is there anything you would like to add?

E. List of attributions for VPN (E) and NAS (E) videos

Below is a list of attributions which must be included in video descriptions of Explainer (animated) videos on YouTube.

1. Question by andrewcaliber
“<https://thenounproject.com/browse/icons/term/question/>”
2. Safety by Adrien Coquet
“<https://thenounproject.com/browse/icons/term/safety/>”
3. Lock by Wouter Buning
“<https://thenounproject.com/browse/icons/term/lock/>”
4. Czech republic by Loudoun Design Co.
“<https://thenounproject.com/browse/icons/term/czech-republic/>”
5. USA Map by Perametade Games
“<https://thenounproject.com/browse/icons/term/usa-map/>”
6. House by Bellowen
“<https://thenounproject.com/browse/icons/term/house/>”
7. Server by nugra
“<https://thenounproject.com/browse/icons/term/server/>”
8. Computer by Patrick Morrison
“<https://thenounproject.com/browse/icons/term/computer/>”
9. Phone by Dave Robson
“<https://thenounproject.com/browse/icons/term/phone/>”
10. Person by Andre
“<https://thenounproject.com/browse/icons/term/person/>”
11. School buildings by Mansion@design
“<https://thenounproject.com/browse/icons/term/buildings/>”
12. Cloud storage by KonKapp
“<https://thenounproject.com/browse/icons/term/cloud-storage/>”
13. Upload Folder by Danan Wijaya
“<https://thenounproject.com/browse/icons/term/upload-folder/>”
14. Download Folder by Danan Wijaya
“<https://thenounproject.com/browse/icons/term/downloadfolder/>”
15. Folder history by Danan Wijaya
“<https://thenounproject.com/browse/icons/term/folder-history/>”
16. Router by sugeng riyanto
“<https://thenounproject.com/browse/icons/term/router/>”
17. Download by kamal az zahra 93
“<https://thenounproject.com/browse/icons/term/download/>”

18. View by The Icon Z
“<https://thenounproject.com/browse/icons/term/view/>”
19. Windows by Manglayang studio
“<https://thenounproject.com/browse/icons/term/windows/>”
20. Android by HideMaru
“<https://thenounproject.com/browse/icons/term/android/>”
21. Magnifying Glass by Jos e Manuel de La a
“<https://thenounproject.com/browse/icons/term/magnifglass/>”
22. Laptop by Yon ten
“<https://thenounproject.com/browse/icons/term/laptop/>”
23. Data Protection by Soremba from
“<https://thenounproject.com/browse/icons/term/data-protection/>”
24. Time by Marie Van den Broeck
“<https://thenounproject.com/browse/icons/term/time/>”
25. Pencil by HideMaru
“<https://thenounproject.com/browse/icons/term/pencil/>”
26. Bin by Luca Santomauro
“<https://thenounproject.com/browse/icons/term/bin/>”
27. Internet by Josy Dom Alexis
“<https://thenounproject.com/browse/icons/term/internet/>”

F. Video descriptions for distribution on the VŠE website

VPN (E)

1. Links to every VPN (T) video.

VPN (T), Windows

1. A link to the VPN (E) video.
2. A link to download Cisco AnyConnect: “vpn.vse.cz”.
3. VŠE server address: “vpn.vse.cz”.

VPN (T), MacOS

1. A link to the VPN (E) video.
2. A link to download Cisco AnyConnect: “vpn.vse.cz”.
3. VŠE server address: “vpn.vse.cz”.

VPN (T), Linux

1. A link to VPN (E) video,
2. VŠE server address: “vpn.vse.cz”,
3. Commands to install a VPN package on different Linux-like operating systems:
 - Ubuntu, Debian, Kali: “sudo apt install network-manager-openconnect-gnome”,
 - Arch, Manjaro, EndeavourOS: “sudo pacman -S networkmanager-openconnect”,
 - Fedora, CentOS Stream, RHEL, AlmaLinux, Rocky Linux: “sudo dnf install NetworkManager-openconnect-gnome”,
 - SUSE: “sudo zypper install NetworkManager-openconnect-gnome”,
 - Clear Linux: “sudo swupd bundle-add NetworkManager-extras”,
 - Gentoo: “sudo emerge -ask net-vpn/networkmanager-openconnect”.

VPN (T), Android

1. A link to the VPN (E) video.
2. A link to download Cisco Secure Client:
“play.google.com/store/apps/details?id=com.cisco.anyconnect.vpn.android.avf”.
3. VŠE server address: “vpn.vse.cz”.

VPN (T), IOS

1. A link to the VPN (E) video.
2. A link to download Cisco Secure Client:
“apps.apple.com/cz/app/cisco-secure-client/id1135064690?l=cs”.
3. VŠE server address: “vpn.vse.cz”.

NAS (E)

1. Links to every NAS (T) video.

NAS (T), Windows

1. VŠE server addresses:
“\\nas1.kme.vse.cz\vyuka” to access Vyuka,
“\\nas1.kme.vse.cz\odevzdavarna” to access Odevzdavarna.
2. Links to every VPN (T) video.

NAS (T), MacOS

1. VŠE server address:
“smb://nas1.kme.vse.cz”.
2. Links to every VPN (T) video.

NAS (T), Linux

1. VŠE server addresses:
“\\nas1.kme.vse.cz\vyuka” to access Vyuka,
“\\nas1.kme.vse.cz\odevzdavarna” to access Odevzdavarna.
2. Links to every VPN (T) video.

G. Contract of the Creation and Use of Artistic Performance Licensing Agreement

Titul, jméno a příjmení: Anna Manukian

Adresa trvalého bydliště: Panušková 1301/4, Praha 4, 140 00

Datum narození: 30.07.1998

e-mail: nickafront@vse.cz

(dále jen "**Objednatel**")

a

Titul, jméno a příjmení: Ivan Růžička

Adresa trvalého bydliště: Škrábkových 883/20, 199 00, Praha 9

Datum narození: 3.11.1965

E-mail: ivan55mov@gmail.com

(dále jen "**Herec**")

uzavírají níže uvedeného dne tuto

SMLOUVU O VYTVOŘENÍ HERECKÉHO VÝKONU A POSKYTNUTÍ LICENCE K JEHO UŽITÍ

Článek 1 - Základní ustanovení

Pracovní název AVD: Co je VPN, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 4 minuty 30 vteřin; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Co je NAS, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 2 minuty 30 vteřin; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Instalace VPN Windows, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 1 minuta 30 vteřin; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Instalace VPN MacOS, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 2 minuty; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Instalace VPN Linux, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 2 minuty; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Instalace VPN iPhone, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 1 minuta 30 vteřin; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Instalace VPN Android, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 2 minuty; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Připojení na NAS Windows, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 1 minuta 30 vteřin; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Připojení na NAS MacOS, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 2 minuty; Specifikace Výkonu: Voiceover.

Pracovní název AVD: Připojení na NAS Linux, režie: Anna Manukian; Účel výroby AVD: výukové a vzdělávací účely; Stopáž AVD: 1 minuta; Specifikace Výkonu: Voiceover.

Doba plnění (předpokládané období výroby AVD): příprava: 1. 09. 2022; natáčení: od 15. 09. 2022 do 15. 11. 2022; postprodukce: od 1. 11. 2022 do 1. 12. 2022.

Článek 2 - Předmět smlouvy

1. Objednatel objednává u Herce, aby dle pokynů Objednatele osobně a výlučně svou vlastní uměleckou činností vytvořil herecký výkon (dále jen "Výkon") pro účely zařazení do audiovizuálních děl specifikovaných v čl. 1 ("AVD"). Požadavky na Výkon a specifikaci AVD dle čl. 1 je Objednatel kdykoli oprávněn měnit, pokud to podstatně nezmění rozsah plnění na straně Herce. Za součást Výkonu bude považována činnost Herce při pořizování fotografií jeho osoby v souvislosti s AVD a při pořízení vizuálu k AVD.
2. Rozsah plnění Herce je dán domluvou stran a zvyklostmi české audiovizuální výroby pro daný typ výkonu a daný druh vyráběného audiovizuálního díla (AVD pro výukové a vzdělávací účely). Herec tak zejména vytvoří Výkon a bude spolupracovat s Objednatelem v průběhu výroby AVD. Konečné rozhodnutí o podobě Výkonu, jakož i o jeho změnách, je vyhrazeno Objednateli. Herec je povinen splnit i veškeré předpoklady nezbytné k vytvoření Výkonu, zejm. nastudovat scénář AVD. Herec je povinen dostavit se na místo vytváření Výkonu vždy včas, připraven, v souladu s dispozicemi dodanými Objednatelem a nikoli pod vlivem návykových látek.
3. Objednatel plánuje vytvoření Výkonu a výrobu AVD na dobu uvedenou v čl. 1, a o termínu vytvoření Výkonu a výrobních termínech AVD bude Herce informovat. Herec je povinen vytvářet Výkon s vynaložením svých schopností a dovedností vlastní uměleckou činností. V rámci doby uvedené v čl. 1. je Objednatel oprávněn výrobní termíny měnit, avšak dobu dle čl. 1. může Objednatel měnit jen z vážných produkčních důvodů při respektování již nasmlouvaných závazků Herce na další období.
4. Herec své povinnosti splní vytvořením Výkonu a jeho schválením Objednatelem. I po schválení Výkonu je Herec povinen poskytovat přiměřenou součinnost Objednateli k případným Objednatelem požadovaným úpravám Výkonu a AVD až do dokončení AVD.
5. Případné vadné plnění je Objednatel povinen vytknout do šesti měsíců od vytvoření Výkonu, je-li zjevné; v opačném případě kdykoli co vyšly vady plnění najevo, nejpozději do šesti měsíců, co se o tom dozvěděl. Herec nesmí zařadit do Výkonu žádné předměty ochrany třetích osob, k jejichž zařazení by Objednatel nedal výslovný souhlas.
6. Herec zajistí, aby Výkon neobsahoval skryté obchodní sdělení, leda k tomu dostane od Objednatele pokyn.

Článek 3 - Poskytnutí práv

1. Herec tímto Objednateli poskytuje bezúplatně oprávnění (licenci) k užití Výkonu zařazením do AVD (použitím v AVD), jakož i k jeho zveřejnění a výhradnímu neomezenému užití všemi způsoby, bez časového, množstevního a teritoriálního omezení, i po zpracování a jiné změně, zcela nebo zčásti, bez povinnosti licenci využít a s možností licenci postoupit nebo poskytnout jiné osobě podlicenci.
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3. Licence dle této smlouvy je poskytována jako výhradní. Herec prohlašuje, že před podpisem této smlouvy neposkytl žádné třetí osobě žádnou výhradní ani nevýhradní licenci k jakémukoli užití Výkonu nebo jeho podstatné části.
4. Herec bere na vědomí, že nositelem práva výrobce zvukově obrazového záznamu AVD, do něž bude Výkon zařazen, bude Objednatel anebo osoba odvozující své oprávnění od práv Objednatele.
5. Licence dle tohoto článku se vztahuje i na všechny materiál zachycující Výkon, avšak nezařazený do konečné podoby AVD, vč. veškerých fotografií pořízených při zaznamenání Výkonu, pořízení vizuálu k AVD a jednotlivých snímků zvukově-obrazového záznamu zachycujících Výkon.
6. Pro případ, že Herec při plnění této smlouvy vytvoří autorské dílo nebo jiný umělecký výkon než je Výkon, nebo jiný předmět ochrany, poskytuje Objednateli výhradní a ničím neomezené svolení k zařazení takového uměleckého výkonu, díla nebo jiného předmětu ochrany do AVD nebo jakéhokoli jiného audiovizuálního díla (i opakovaně), včetně oprávnění je neomezeně užit ve stejném rozsahu jako Výkon dle tohoto článku, a to bezúplatně, přičemž se zavazuje sám je po dobu trvání licence nezařadit do žádného audiovizuálního díla ani neposkytnout jinému oprávnění k jakémukoli jejich užití.
7. Herec je povinen se zdržet výkonu práva AVD užit s výjimkou oprávnění užití úryvků z AVD autorem k prezentaci své činnosti (self promotion). Užití úryvků z AVD pro self promotion nepřesáhne v žádném jednotlivém případě (i v případě po stříhových úpravách) 10% finální stopáže AVD.
8. Herec poskytuje Objednateli neodvolatelné oprávnění (licenci) k užití svých osobnostních atributů, zejména jména a podobizny, po dobu 50 let, a to pro účely související s Výkonem a AVD, včetně užití fotografií Herce pořízených v souvislosti s AVD, vizuálem AVD a Výkonem (ať již samostatně nebo jako jednotlivých snímků zvukově-obrazového záznamu), výroby tzv.

filmu o filmu a podobných snímků ze zákulisí, není-li takové užití kryto zákonnou licencí. Herec prohlašuje, že netrvá na utajování svého pseudonymu, pokud užívá pseudonym.

Článek 4 - Další ujednání

1. Herec vytvoří Výkon bezúplatně a poskytuje práva k němu v rozsahu dle čl. 3 rovněž bezúplatně.
2. Tato smlouva se uzavírá co do poskytnutých práv na dobu určitou – na celou dobu trvání jejich právní ochrany, není-li ohledně některých práv v čl. 3 této smlouvy uvedeno jinak.
3. Pokud se Herec dostane do prodlení s plněním svých povinností, má Objednatel přesto zájem i na opožděném plnění Herce a závazek počátkem prodlení Herce nezaniká (což nevylučuje právo Objednatel od smlouvy odstoupit).
4. Poruší-li Herec některou ze svých povinností dle této smlouvy, má Objednatel právo od této smlouvy bez dalšího odstoupit, a to buď zcela, nebo zčásti (v rozsahu vadného anebo neposkytnutého plnění).
5. Dojde-li k zastavení výroby AVD před dokončením a předáním, je Objednatel oprávněn od této smlouvy odstoupit v rozsahu týkajícím se nedokončené části Výkonu.

Článek 5 - Závěrečná ustanovení

1. Herec nesmí bez pokynu Objednatel při vytváření Výkonu a výrobě AVD pořizovat vlastní záznamy, informovat na sociálních sítích ani jinak o vytváření Výkonu a výrobě AVD a jejich stavu a dalších detailech Výkonu a AVD (jako například o zápletky AVD), fotografovat při výrobě (zejména při natáčení či v zákulisí) apod. Informace, které nejsou veřejné a o nichž se při plnění smlouvy Herec dozví, musí uchovávat v důvěrnosti, a to bez časového omezení.
2. Tato smlouva se vyhotovuje ve 2 originálních stejnopisech, po 1 pro každou ze smluvních stran. Veškeré spory z této smlouvy se smluvní strany zavazují řešit především smírnou cestou, jinak je dána pravomoc soudů České republiky a bude rozhodováno podle zákonů České republiky.
3. Smluvní strany po přečtení smlouvy prohlašují, že souhlasí s jejím obsahem, že byla sepsána na základě pravdivých údajů a jejich pravé vůle. Na důkaz toho připojují své podpisy.

V _____ dne _____

V _____ dne _____

Objednatel: _____

Herec: _____

Objednatel upozorňuje Herce na skutečnost, že pro účely plnění této smlouvy musí zpracovávat jeho osobní údaje. Podrobné informace o tom, jaké osobní údaje jsou zpracovávány a v jakém rozsahu, jakož i poučení Herce o jeho právech s tím souvisejících nalezne Herec na internetových stránkách Objednatel (<https://www.vse.cz/informace-o-vse/informace-a-predpisy/gdpr/>). Poučení jsem porozuměl a s jeho zněním na stránkách Objednatel jsem se seznámil:

Herec: _____

H. URL address of the produced videos

The videos described in this thesis can be found in this YouTube playlist. ([clickable link](#))